

Decoding protein phosphorylation during oocyte meiotic divisions using phosphoproteomics

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Leonid Peshkin , Enrico Maria Daldello, Elizabeth Van Itallie, Matthew Sonnett, Johannes Kreuzer, Wilhelm Haas, Marc W Kirschner, Catherine Jessus

Systems Biology Department, Harvard Medical School, Boston, United States • Sorbonne Université, CNRS, Laboratoire de Biologie du Développement - Institut de Biologie Paris Seine, LBD - IBPS, Paris, France

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eLife Assessment

This **important** paper describes a comprehensive quantitative phospho-proteomic analysis of the meiotic progression of *Xenopus* oocytes. Using time-resolved proteomic analyses, the authors provide insights into changes in protein levels and phosphorylation states to an unprecedented depth, quality, and quantitative detail. The key findings are **compelling** and offer a helpful resource for the scientific community.

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Abstract

Oocyte meiotic divisions represent a critical process in sexual reproduction, as a diploid non-dividing oocyte is transformed into a haploid fertilizable egg, as a prelude for the subsequent embryonic divisions and differentiation. Although cell differentiation and proliferation are governed by transcription, oocyte maturation and early embryonic divisions depend entirely on changes in protein abundance and post-translational modifications. Here, we analyze the abundance and phosphorylation of proteins during *Xenopus* oocyte meiotic maturation. We reveal significant shifts in protein stability, related to spindle assembly, DNA replication and RNA-binding. Our analysis pinpoints broad changes in phosphorylation correlating with key cytological meiotic milestones, noteworthy changes in membrane trafficking, nuclear envelope disassembly and modifications in microtubule dynamics. Additionally, specific phosphorylation events target regulators of protein translation, Cdk1 and the Mos/MAPK pathway, thereby providing insight into the dynamics of Cdk1 activity, as related to the meiotic cell cycle. This study sheds light on the orchestration of protein dynamics and phosphorylation events during oocyte meiotic divisions, providing a rich resource for understanding the molecular pathways orchestrating meiotic progression in the frog, and most likely applicable to other vertebrate species.

Introduction

In mature females, the oocyte undergoes a long period of growth, during which meiosis is arrested at prophase I. In the *Xenopus* oocyte, transcription is switched off when the oocyte reaches the end of its growth phase (Dumont, 1972). The full-grown oocyte is transcriptionally silent and equipped with a stock of mRNAs and proteins that will support without transcription three critical embryonic processes: first, two meiotic divisions that transform the oocyte into a fertilizable haploid egg; second, the process of fertilization; and third, 12 embryonic divisions, after which transcription is initiated.

The transformation of a fully-grown oocyte arrested in prophase into a fertilizable cell, arrested in metaphase of the second meiotic division is a process called meiotic maturation. Meiotic maturation is triggered by progesterone, secreted by the follicle cells surrounding the oocyte. The secretion of progesterone is stimulated by Luteinizing Hormone produced by the pituitary. Progesterone initiates a molecular signaling cascade that lasts 3 to 4 hours and occurs with little change in oocyte morphology. Notably, this cascade leads to the activation of the Cdk1-Cyclin B kinase, the universal inducer of cell division in eukaryotes. Cdk1 phosphorylates many substrates that promote meiotic nuclear events: nuclear envelope breakdown (NEBD), the formation of the first meiotic spindle marking the metaphase I (MI), the completion of the first meiotic division with expulsion of a polar body, and formation of the second meiotic spindle (Figure 1). As in all vertebrates, the frog oocyte arrests a second time in metaphase II (MII), until fertilization stimulates the completion of the second meiotic division. The entire process of oocyte maturation occurs in the absence of transcription. The regulation of the two meiotic divisions and the 13 embryonic divisions that follow are thought to be largely regulated by changes in protein abundance and phosphorylation/dephosphorylation.

The prophase-arrested oocyte contains a store of Cdk1-Cyclin B complexes that are kept inhibited by Cdk1 phosphorylation at Y15 and T14, which are substrates of the kinase, Myt1 (Mueller et al, 1995). In all vertebrates, the prophase arrest is maintained by high levels of cAMP and PKA (cAMP-activated protein kinase) activity (Figure 1). The release of this prophase block is triggered by a drop in the levels of cAMP and the consequent inhibition of PKA, which occurs within 60 min after progesterone stimulation. The identity of the critical PKA substrates whose dephosphorylation is thought to induce the pathway leading to Cdk1 activation is still unknown, with the exception of Arpp19 whose mechanism of action is not completely understood (Dupré et al, 2014; Santoni et al, 2024). Nevertheless, a drop in PKA activity leads to the synthesis of new proteins from cytoplasmic stockpiles of mRNAs as well as increases in protein level through regulation of the ubiquitination machinery (Santoni et al, 2024). Among the accumulated proteins are Cyclin B1, the best-known activator of Cdk1 in mitotic cells, RINGO/Speedy (Ferby et al, 1999; Lenormand, 1999) and Mos, a kinase specific to the oocytes (Figure 1). The newly synthesized B1-Cyclins bind to monomeric Cdk1 to form a small pool of active complexes that evade the inhibition of Myt1 (Hochegger et al, 2001; Gaffré et al, 2011). Subsequently, this small amount of active Cdk1 initiates a complex network of feedback loops, involving many kinases and phosphatases, thereby creating an auto-amplification loop (Jesus, 2010) (Figure 1). Within this loop, the activation of the Cdc25 phosphatase that dephosphorylates Cdk1 and the inhibition of the PP2A phosphatase that counteracts Cdk1, lead to a rapid and full activation of Cdk1-Cyclin B (Jesus, 2010; Lemonnier et al, 2020). Another important player in this positive feedback loop is Mos, whose translation is induced by progesterone and which accumulates at the time of Cdk1 activation (Sagata et al, 1988, 1989a; Frank-Vaillant et al, 1999). Mos activates Erk1/2 (also known as MAPK) indirectly, which modulates the core regulators of Cdk1 (Haccard & Jesus, 2006) (Figure 1). Cdk1 and the kinases activated under its control (Mos/MAPK, Aurora-A, Plk1, etc.) trigger a second wave of protein translation and accumulation, as well as mediating the structural changes of cell division: NEBD, chromosome condensation and formation of the

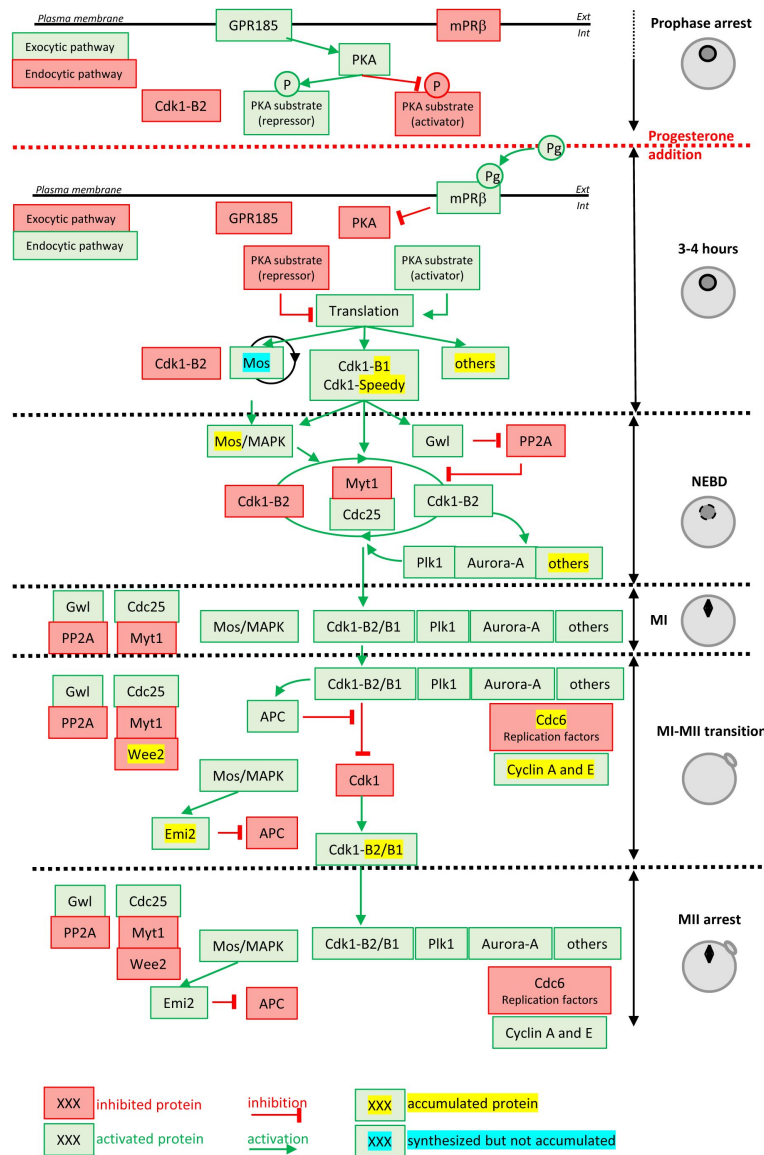


Figure 1.

Molecular events coordinating *Xenopus laevis* oocyte meiotic divisions.

From top to bottom: The prophase arrest is maintained by high PKA activity due to the constitutively expressed GPR185 receptor. The exocytic pathway is active while the endocytic pathway is downregulated. Cdk1-Cyclin B2 complexes are present under an inactive state due to Cdk1 phosphorylation. Progesterone activates the mPR β receptor, GPR185 is inactivated, the exocytosis and endocytosis pathways are inverted. PKA is inactivated and promotes translation and/or accumulation of new proteins, among them Mos (that does not accumulate due to its instability) and Cyclin B1 (due to its stabilization). New Cyclin B1 proteins associate with Cdk1 and form a starter amount of Cdk1 activity. This Cdk1 starter stabilizes Mos and promotes Cdk1-Cyclin B2 activation through the regulation of Myt1, Cdc25, Gwl and PP2A. The autoamplification loop is launched, whereby more Cdk1 is activated, more it activates kinases as Mos-MAPK, Aurora-A and Plk1, all of them contributing with Cdk1 to regulate Myt1 and Cdc25, hence accelerating Cdk1 activation. Phosphorylated substrates trigger NEBD and the MI spindle assembly as well as protein translation. APC is activated under the control of Cdk1 and leads to Cyclin B degradation. The Cdk1 activity decreases and allows anaphase I and expulsion of the 1st polar body. Under the Mos/MAPK control, Emi2/Erp1 inactivates APC, halting Cyclin B degradation. The re-accumulation of Cyclin B increases Cdk1 activity that triggers the formation of the MII spindle, avoiding the assembly of a nucleus and DNA replication. APC remains inhibited by Emi2/Erp1, what ensures the oocyte arrest in MII. Color code: red: inactivated proteins; green: activated proteins; red arrow: inhibition; green arrow: activation; yellow highlight: accumulated protein; blue highlight: protein synthesized but not accumulated.

Meiosis I spindle (Jesus, 2010 [↗](#); Santoni et al, 2024 [↗](#)). They also activate the anaphase-promoting complex (APC), an E3 ubiquitin ligase, leading to the degradation of both Cyclin B and securin, thus enabling entry into anaphase I, chromosome separation and the expulsion of the 1st polar body (Figure 1 [↗](#)). The decrease in Cdk1 activity is rapidly followed by the inhibition of APC, which then allows the accumulation of B-Cyclins and increase of Cdk1 activity (Figure 1 [↗](#)). These events trigger the entry into metaphase II and the formation of the MII spindle. In all vertebrates, oocytes arrest in the middle of the meiotic division as a result of the stabilization of cyclin B, resulting from APC inhibition by the Emi2/Erp1 (Jesus, 2010 [↗](#)) (Figure 1 [↗](#)).

The burst of phosphorylation, which occurs at NEBD when Cdk1 is activated, targets proteins belonging to three categories (Maller et al, 1977 [↗](#)): first, regulators of translation and/or degradation of proteins that are essential for the progression of meiotic divisions; second, proteins directly involved in the intracellular reorganization of the oocyte, such as the components of meiotic spindles, the nuclear envelope or other cytoskeletal elements; third, proteins that do not have a role in meiotic divisions but regulate fertilization and early embryonic development. Notably, the sperm does not contribute any proteins or RNA except for its genome and its two centrioles. With fertilization there is extrusion of the 2nd polar body, the fusion of the male and female pronuclei. Within a few minutes after fertilization, the first S-phase starts, followed by the first 12 embryonic division cycles, all occurring in the absence of transcription (Blitz & Cho, 2021 [↗](#)). Thus, these events depend exclusively on maternal proteins and mRNA.

Phosphoregulation is a central and conserved mechanism that enables remodeling of the oocyte proteome and supports cell division. It prepares the oocyte to transform into the egg and undertake development. In the absence transcription our knowledge of the dynamic patterns of phosphorylation of thousands of oocyte phosphosites represents an essential tool for study of oogenesis, embryogenesis, especially the unique meiotic and mitotic processes. Interestingly, two recent studies based on phosphoproteomics have revealed new insights into how phosphorylation dynamics regulate meiotic divisions in yeast (Celebic et al, 2024 [↗](#); Koch et al, 2024 [↗](#)) and mouse (Sun et al, 2024 [↗](#)). We now extend this analysis to the physiological process of oogenesis specific to non-mammalian metazoans. The study of mitotic processes in somatic cells, which are short in duration, requires exquisite synchronization, which is difficult. Although previous studies of protein phosphorylation during mitosis have revealed the identity of numerous players of cell division, especially those controlling the structural reorganization of the dividing cell and its checkpoints, they have not been able to fully elucidate the epistatic relationships between kinases and phosphatases, which are nonetheless crucial to understanding the temporal sequences regulating the various events of mitosis. This weakness is due to the fact that cell proliferation is a continuous phenomenon, without physiological arrest. Its duration varies from one cell to another within the same population. The necessity of using synchronizing agents induces non-physiological arrests, which activate checkpoints, leading to the resumption of the cell cycle not comparable to that of untreated cells. Moreover, these studies often use transformed cell lines with accumulation of mutations that makes it hard to compare with cells in a physiological cellular context. The exceptional features of oocyte maturation for experimental studies of mitosis and meiosis are: 1) the synchronicity of the process in the oocyte, initiated by a physiological signal, progesterone, 2) the natural oocyte arrests in oocyte maturation at either prophase I (equivalent to a late G2-arrest) or at MII (M-phase arrest), 3) and the large size of the oocyte (1.2 mm in diameter) and its high protein content (30 µg soluble proteins per cell), offering the rare opportunity of single-cell proteomics. In addition, the *Xenopus* model offers great assets. From a technical perspective, in the mouse, the small number of oocytes and their low protein content limit proteomic approaches (30,000 oocytes isolated from 950 mice were required for the phosphoproteomic profiling by (Sun et al, 2024 [↗](#))). Furthermore, the resumption of meiotic divisions in mouse obeys a particular regulation: unlike other mammalian models (including the human species) and other vertebrates whose entry into meiosis I depends strictly on protein translation, it does not require the synthesis of new proteins, making it a somewhat marginal

model (Meneau et al, 2020 [link](#)). For these reasons, we performed time-resolved proteomics and phosphoproteomics in *Xenopus* oocytes from prophase I through the MII arrest of the unfertilized egg.

In this study, we have analyzed the changes in the phosphorylation in relation to three categories of events: specific periods of meiotic maturation, activities of the master regulators of meiosis and peculiar cellular features of meiosis. Our analysis confirms the high quality of our phosphoproteome by the detection of multiple phosphorylation events with previously-documented functional effects on meiotic divisions or early development. Through the detection of many new regulated phosphosites, it provides a rich pool of candidate proteins for multiple avenues of investigation of previously uncharacterized important players of the oocyte to embryo transition.

Results

The temporal resolution of the events of meiotic divisions

To determine both proteome and phosphoproteome of *Xenopus* oocytes during meiosis resumption, meiotic maturation was triggered by progesterone, and oocytes were collected at regular intervals after progesterone exposure (Movie EV1, Methods): prophase oocytes (PRO) at the time of progesterone addition, then 2 hours after progesterone treatment, with hourly timepoints past that, up to 9 hours post-treatment. Groups of ten or fifty oocytes were collected at each timepoint for protein and phospho - proteomics measurements respectively. After oocyte lysis, digestion into peptides and phospho-peptide enrichment in the case of phosphoproteomics analysis, samples were processed for TMT-MS3 LC/MS (Methods).

To correlate the time-points of this experiment with the cytological events known to occur during meiosis resumption, a set of proteins whose accumulation had been extensively studied during *Xenopus* oocyte meiosis progression were used as markers. One of the first events following progesterone stimulation is the accumulation of Cyclin B1 that begins before and independently of Cdk1 activation (Santoni et al, 2024 [link](#); Frank-Vaillant et al, 1999 [link](#)). In our experiments, Cyclin B1 accumulation starts at about 2 hours and increases linearly until MII arrest at 6.5 hours (**Figure 2A** [link](#)). It has been reported that Mos translation is stimulated by progesterone concomitant with the accumulation of Cyclin B1; however the Mos protein does not accumulate before NEBD because it turns over rapidly (Santoni et al, 2024 [link](#); Frank-Vaillant et al, 1999 [link](#)). Another critical well documented event that occurs downstream to Cdk1 activation is the degradation of CPEB1 (Santoni et al, 2024 [link](#); Mendez et al, 2002 [link](#)), an RNA-binding protein whose degradation activates protein translation in MI. Hence, Mos accumulation and the degradation of CPEB1 are well established markers for the timing of NEBD. After NEBD, activation of the anaphase-promoting complex (APC) leads to the ubiquitination and degradation of Cyclin B2, which marks the entry into anaphase I. During the MI-MII transition, multiple additional events take place: Cyclin B3 is degraded (Bouftas et al, 2022 [link](#)), Cdc6 (Lemaître et al, 2002 [link](#); Whitmire et al, 2002 [link](#); Daldello et al, 2015 [link](#)), Wee2 (Nakajo et al, 2000 [link](#)) and Cyclin E (Rempel et al, 1995 [link](#)) accumulate, reaching their maximal levels in MII. These four events mark the entry into MII. Based on these markers, the timing of cytological events in our experiment is as follows (**Figure 2A** [link](#)): NEBD and MI occur between 3 and 4 hours, the MI-MII transition takes place between 5 and 6 hours, and entry and arrest in MII are at 7 hours. This time course is in agreement with the cytological previously described events (Huchon et al, 1981 [link](#); Gard, 1992 [link](#)). A physiological measure of MII arrest can be ascertained by the ability of the oocytes to be activated by an electric shock, a property that is acquired only in MII (Movie EV2).

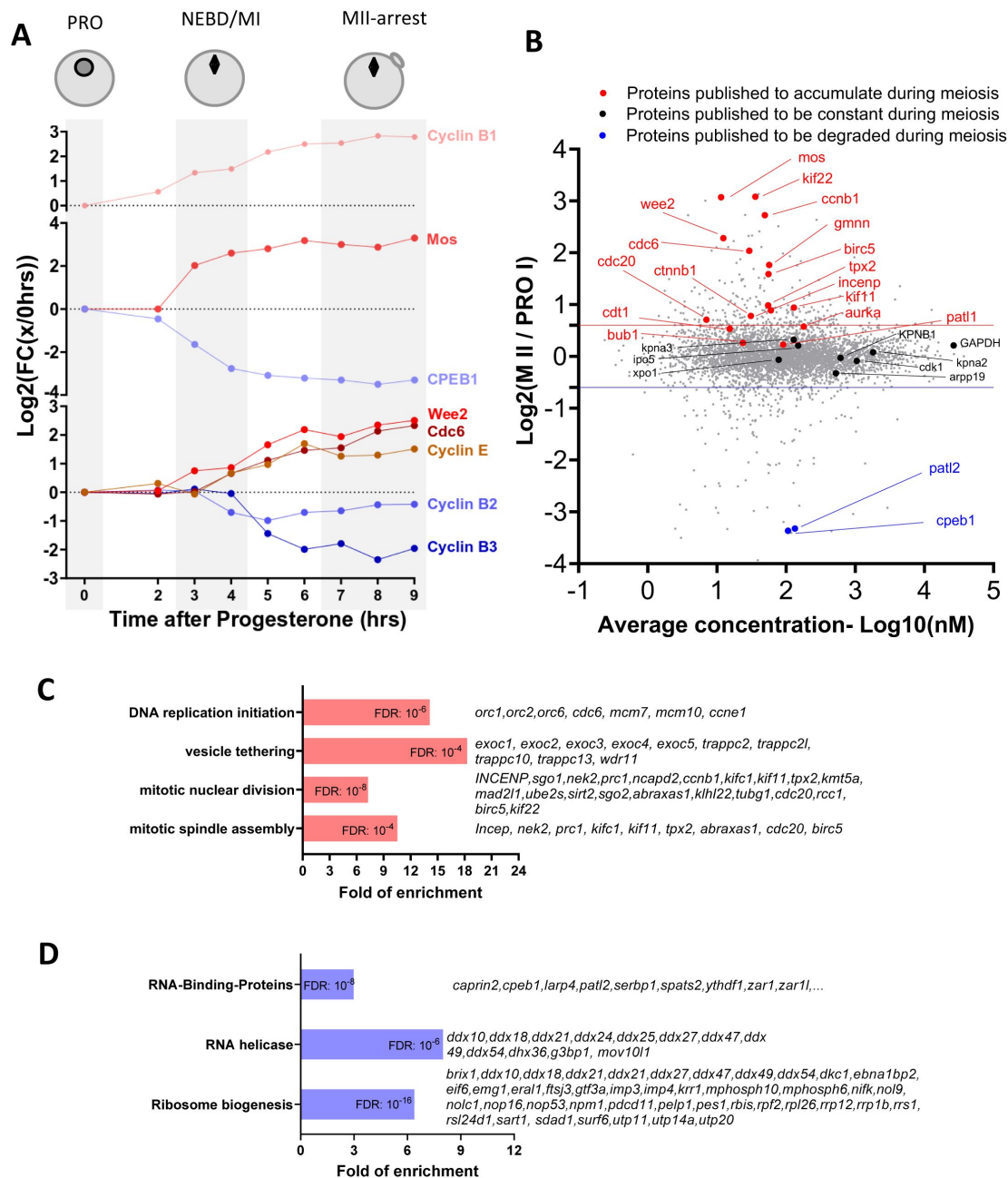


Figure 2.

Changes in the protein levels during meiotic divisions.

(A) The relative changes in the concentration of proteins whose accumulation/degradation has been reported to correlate with the critical stages of *Xenopus* meiotic divisions have been plotted. (B) Changes of protein stoichiometry during meiotic maturation. The log2 fold changes between the concentration of each protein in PRO-arrested oocytes and MII-arrested oocytes was plotted against the average concentration during meiotic divisions. The red line marks the 1.5-fold change used as a threshold to identify accumulating proteins. The blue line marks the 0.75-fold change used as threshold to distinguish proteins that are degraded during meiosis. Proteins, whose changes in homeostasis during meiotic divisions are documented in the literature, are marked with the following color code: red, for proteins accumulating; black, for proteins which are expressed at a stable level; and blue, for proteins that are degraded. (C-D) Gene ontology analysis of the biological processes enriched among the proteins found to accumulate (in red, panel C) or degraded (in blue, panel D) during meiosis. The statistical significance of the enrichment is expressed as false discovery rate (FDR). A representative subset of the proteins belonging to each group is displayed.

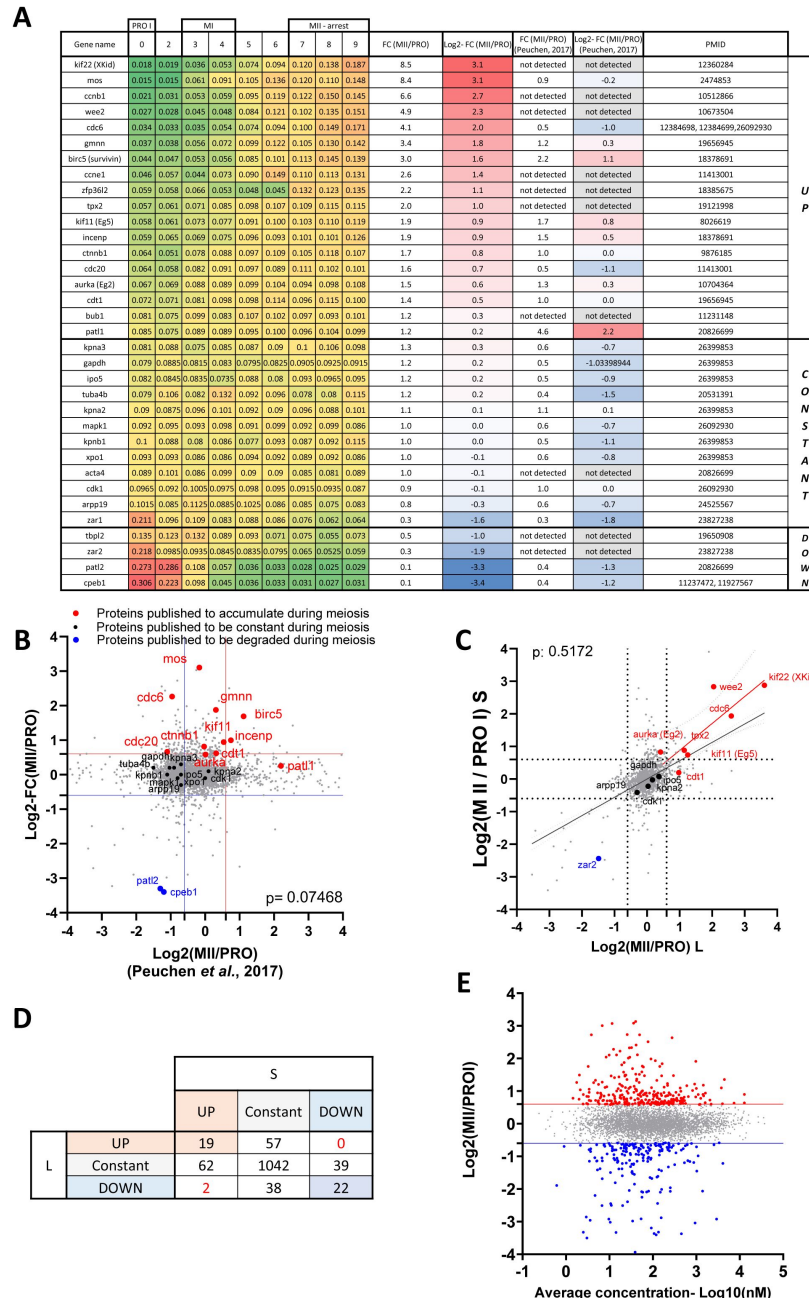


Figure 2-figure supplement 3.

Validation of the proteomic dataset.

(A) Table summarizing proteins whose changes in protein level have been documented using Western blot. The PMID of the original publication displaying the Western blot is reported. The changes in abundance detected in our dataset are color-coded with a scale from green to red. Log2 fold changes between prophase and metaphase II from our dataset are color-coded with a scale from red to blue and compared to another previously published dataset (Peuchen et al., 2017). (B) Proteome-wide comparison of our dataset with (Peuchen et al., 2017). Proteins whose abundance was previously published are color-coded in red for the ones that accumulate, black for the constant, and blue for the ones that are degraded. (C) Changes in protein abundance between prophase and metaphase II were calculated for each protein where we could distinguish the .S and .L isoforms and plotted. The Pearson correlation between the isoforms was calculated with Prism. (D) Table summarizing the comparison between .S and .L isoforms presented in panel C. (E) Correlation plot between the changes in protein abundance occurring during the prophase to metaphase II transition and the average concentration of the proteins during meiotic divisions.

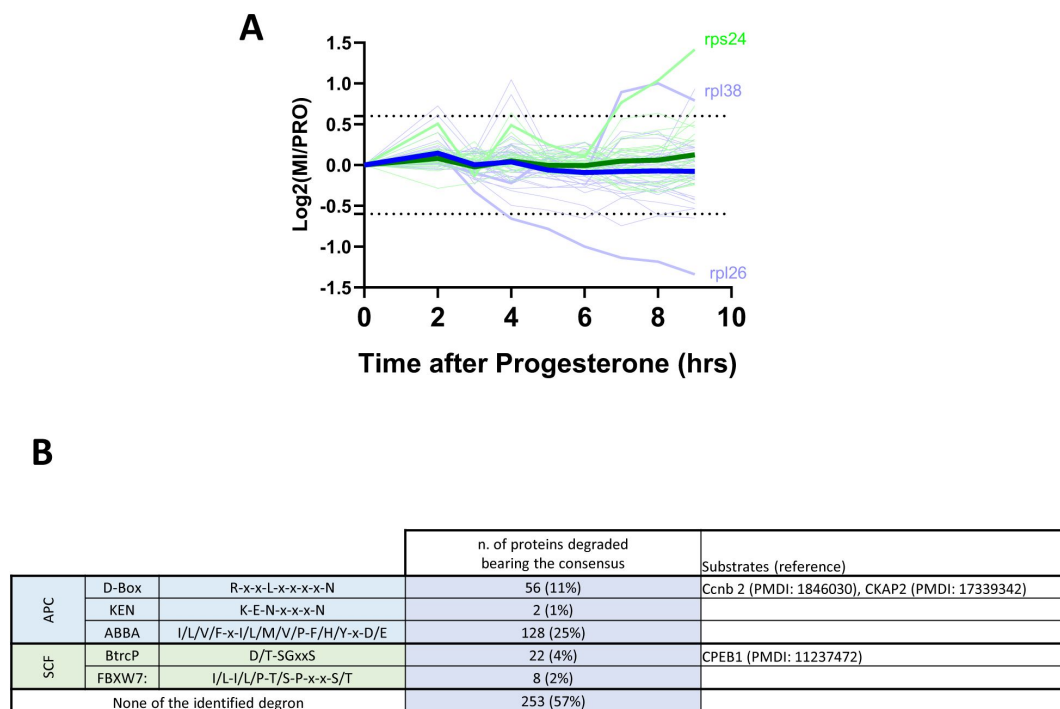


Figure 2-figure supplement 4.

Protein degradation during meiotic maturation: ribosomal proteins, E3 ligases targets.

(A) Changes in protein abundance of ribosomal proteins between prophase (PRO) and metaphase II (MII) are plotted. Dashed lines represent the threshold used for defining the differentially expressed proteins. (B) Table summarizing the different degron consensus motifs identified and the number of proteins in our dataset bearing these motifs. Examples of proteins bearing different degrons are listed along with supporting literature.

Validation of the quantitative proteomic data

The quality of oocytes resuming meiosis is also confirmed by the dynamics of protein accumulation and degradation that occur during meiotic progression (**Figure 2B** [↗](#)). A set of 34 proteins, whose changes in concentration have been previously studied during meiotic divisions was used to validate the quality of our quantitative proteomics data (**Figure 2B-figure supplement 3A** [↗](#)). Our proteome correctly classifies 15 out of 18 proteins accumulating during *Xenopus* meiosis resumption ($FC(MII/PRO) > 1.5$), 11 out of 12 proteins whose concentration is constant ($0.75 < FC(MII/PRO) < 1.5$), and 4 out of 4 proteins whose accumulation was reported to decrease ($FC(MII/PRO) < 0.75$) (**Figure 2B-figure supplement 3A** [↗](#)). The concordance between our dataset (30/34 proteins) with the changes in the protein levels reported in the literature is substantially higher than in a previously published system-wide proteomic dataset from *Xenopus* oocytes where only 7 out of these 34 proteins were correctly measured (Peuchen et al, 2017 [↗](#)) (**Figure 2-figure supplement 3A-B** [↗](#)). This also may explain why changes in protein concentration reported in that study poorly correlate with the one measured in our proteomic study (**Figure 2-figure supplement 3B** [↗](#)). The high quality of the new proteomic data is also confirmed by the strong correlation of the translational pattern of *Xenopus* allo-alleles (**Figure 2-figure supplement 3C-D** [↗](#)), which were derived by genome endoduplication (Session et al, 2016 [↗](#)).

Protein homeostasis during meiotic divisions

Absolute concentrations of proteins during meiotic maturation

The absolute concentration of 7974 proteins identified in our proteome dataset was calculated (See Appendix 1). We then performed a qualitative comparison of average oocyte abundance of proteins belonging to distinct functional groups (see Appendix 2). As expected, the less abundant subgroups include signaling molecules and transcription factors. Perhaps of more surprise, E3 enzymes and kinases are not abundant. At the other end of the spectrum are glycolytic and tricarboxylic acid cycle enzymes, as well as proteasome and ribosome components. These observations are expected because the oocyte has a stock of nutrient molecules and all the enzymes linked to their metabolism, which are used in viviparous animals throughout embryonic and larval development. For instance, the glycolytic enzyme GAPDH that catalyzes an important energy-yielding step in carbohydrate metabolism, is a very abundant protein in the oocyte (average concentration of between 8.88 μ M and 17.83 μ M). The ribogenesis program, which takes place during the oocyte growth from stage III to stage VI, before meiosis resumption, allows the accumulation of very large quantities of ribosomal RNAs and proteins that are used during embryonic development, until the swimming tadpole stage (Woodland, 1974 [↗](#)). Similarly, the components of the DNA replication machinery are quite abundant, probably in anticipation of the post-fertilization cell cycles. All these data are therefore consistent with biological understanding of their roles during embryonic development after post-fertilization (Peshkin et al, 2015 [↗](#)).

Temporal profiles of protein levels

We clustered the relative data for protein abundance across subsequent hourly timepoints post-progesterone stimulation (see Methods). As illustrated in Appendix 3A, the majority of the 60 expression profiles are flat, demonstrating that most (over 80%) proteins are stable during meiotic maturation. Our proteomic dataset allowed to identify 12% of proteins affected by changes in concentration during meiotic progression by applying the same fold change threshold: 413 proteins that accumulate and 562 proteins whose concentration decreases (**Figure 2B-figure supplement 3E** [↗](#); see Appendix 4). Although these changes only concern a minority of proteins,

they are critical for the meiotic process. Since transcription is silent during this period, these documented changes must result from the regulation of translation or/and degradation (Meneau et al, 2020 [DOI](#)).

Protein accumulation: the machinery of cell division and DNA replication

Gene ontology analysis was used to characterize the functional significance of the 413 proteins whose level increases during meiosis. Proteins involved in spindle assembly and mitotic division were significantly enriched (**Figure 2C** [DOI](#)). Among these important regulators are the Targeting-protein-for-Xklp2 (Tpx2) and Protein-regulator-of- cytokinesis 1 (Prc1) that also accumulate during meiotic maturation in mouse oocytes (Brunet et al, 2008 [DOI](#); Li et al, 2021 [DOI](#)). We also find that the centrosomal maturation factor, SSX2IP, a plus-end motor kinesin, Kif11/Eg5, and a minus-end one, Kifc1 (Houliston et al, 1994 [DOI](#)), also accumulate significantly during meiotic maturation. Three of these proteins, Tpx2 and the two kinesins, play an important role in the assembly of both acentrosomal meiotic spindles and centrosomal mitotic spindles (Gruss, 2018 [DOI](#); Kufer et al, 2002 [DOI](#); Ems-McClung et al, 2020 [DOI](#); Miller et al, 2019 [DOI](#)). Interestingly, the prophase oocyte is equipped with most of the proteins essential for the formation of division spindles. Hence these new results suggest that increases in the relative level of some components is required to support the meiotic and first rapid embryonic divisions.

Another protein category found to increase is the machinery of DNA replication initiation (**Figure 2C** [DOI](#)). Indeed, oocytes accumulate the components of the DNA replication machinery during meiosis resumption to support the 12 rapid rounds of cell cycles occurring during early embryogenesis, all of which occur in the absence of transcription. In *Xenopus*, Cdc6 is a component of the pre-replicative (pre-RC) complex that is not detectable in the prophase-arrested oocyte by western blot (Lemaître et al, 2002 [DOI](#); Whitmire et al, 2002 [DOI](#); Daldello et al, 2015 [DOI](#)). Cdc6 accumulation occurs during the MI-MII transition to confer oocytes the competence to replicate DNA (Lemaître et al, 2002 [DOI](#); Whitmire et al, 2002 [DOI](#); Daldello et al, 2015 [DOI](#)) (**Figure 1** [DOI](#)). Translational and then post-translational mechanisms targeting Cdc6 prevent the pre- RC from functioning between the two meiotic divisions, thus ensuring the production of a haploid gamete. Our proteomic analysis shows that, additional pre-RC proteins, as ORC1, ORC2 and ORC6 accumulate during meiotic maturation, revealing a strong co-regulation of the components of the pre-RC.

Another process controlled by highly enriched proteins is vesicle tethering, which includes the family of exocyst proteins (**Figure 2C** [DOI](#)). Interestingly, exocyst proteins are involved in the completion of cell division during the secretory-vesicle-mediated abscission (Gromley et al, 2005 [DOI](#)). The accumulation of these proteins during meiotic maturation could be important to support the extrusion of the polar bodies, as already suggested in *C. elegans* (Kumar et al, 2019 [DOI](#)) and the rapid cell divisions sustaining early embryogenesis.

Protein degradation: RNA-binding proteins (RBPs)

Gene ontology analysis of the 562 proteins whose concentration decreases during meiosis reveals that 116 (21 %) are annotated to bind RNA. Among them, a large portion are RNA helicases (**Figure 2D** [DOI](#)). Among these RBPs, another important group are proteins that are involved in ribosome biogenesis. The ribogenesis program takes place during the oocyte growth, allowing the accumulation of ribosomal RNAs and proteins that are used during embryonic development (Woodland, 1974 [DOI](#)). This process is downregulated during meiotic maturation (Hyman & Wormington, 1988 [DOI](#); Chen et al, 2011 [DOI](#)). The decrease in the concentration of proteins involved in rRNA synthesis and processing does not result in a general decrease of ribosomal proteins that are stable during meiotic divisions (**Figure 2-figure supplement 4A** [DOI](#)). Halting ribogenesis may allow the oocyte to re-allocate its energy resources to the production of other cellular components required for meiotic divisions and embryogenesis. Intriguingly, Rpl26 (Ribosomal protein L26) is the only ribosomal protein detected to decrease during meiotic divisions (**Figure 2-figure**

supplement 4A [↗](#)). Since this protein is known to be dispensable for ribosome formation in yeast (Babiano et al, 2012 [↗](#); Warner & McIntosh, 2009 [↗](#); Takagi et al, 2005 [↗](#)), the distinct regulation of Rpl26 as compared to the other Rpls suggests that Rpl26 might have extra-ribosomal functions. This hypothesis is reinforced by its specific modification by the ubiquitin-like UFMylation modification (Walczak et al, 2019 [↗](#)) and its role in the regulation of protein stability (Zhang et al, 2016 [↗](#)). Another subgroup of RBPs that decrease during meiotic divisions is involved in the repression of translation during prophase arrest, such as CPEB1 (Mendez et al, 2002 [↗](#); Reverte et al, 2001 [↗](#); Piqué et al, 2008 [↗](#)), PATL2 (Nakamura et al, 2010 [↗](#); Cao et al, 2021 [↗](#); Christou-Kent et al, 2018 [↗](#)) and Zar11 (Zar2) (Charlesworth et al, 2012 [↗](#); Yamamoto et al, 2013 [↗](#); Rong et al, 2019 [↗](#)). Remodeling the repertoire of RBPs in the oocyte could be a general mechanism to regulate translation during meiotic divisions. Among these RBPs is Serbp1, which in mouse is involved in the stabilization of mRNA involved in meiosis resumption (Chew et al, 2013 [↗](#)) and in ribosome hibernation (Leesch et al, 2023 [↗](#)), as well as Caprin2, whose degradation is also observed during meiotic maturation of human oocytes (Virant-Klun et al, 2016 [↗](#)). Caprin2 is a RBP with unknown function and highly enriched in the oocyte Balbiani body (Boke et al, 2016 [↗](#)), a non-membrane compartment specific to the early diplotene oocytes, which contains mitochondria, RNAs and endoplasmic reticulum. The functions of this organelle are not clear, although it is strongly correlated in *Xenopus* to the establishment of oocyte polarity and to the localization of maternal determinants (Jamieson-Lucy & Mullins, 2019 [↗](#)). In *Xenopus*, it disperses at the onset of vitellogenesis, leaving a wedge-shaped cytoplasmic region rich in mitochondria (Wilk et al, 2005 [↗](#)) and promotes the formation of RNP condensates in the vegetal cortex of the oocyte (Yang et al, 2022 [↗](#)). Interestingly, Rbpms2, another regulator of the Balbiani body (Kaufman et al, 2018 [↗](#)), is strongly degraded during meiotic maturation, highlighting how dynamic during meiosis is the composition of this transient compartment specific to the oocyte.

Interestingly, distinct E3 ubiquitin ligases, mainly APC and SCF (Skp, Cullin, F-box containing complex) have been shown to be active at different time periods of meiosis resumption in *Xenopus* oocytes (Kinterová et al, 2022 [↗](#)). 186 proteins (37%) identified as decreasing in our proteome include in their sequences one of the APC degrons (**Figure 2-figure supplement 4B** [↗](#)). This group of proteins includes two known APC targets, Cyclin B2 (Glötzer et al, 1991 [↗](#)) and Securin (Cohen-Fix et al, 1996 [↗](#)). Additionally, 30 proteins (6%) whose concentration decreases in the oocytes have one SCF degron in their sequence (**Figure 2-figure supplement 4B** [↗](#)), including CPEB1 (Reverte et al, 2001 [↗](#)) and Cdc6 (Daldello et al, 2015 [↗](#)), two characterized SCF-targets. Interestingly, 253 proteins (57%) of proteins that decrease during meiotic divisions do not bear in their sequence any of the degrons for the ubiquitination systems identified in *Xenopus* oocytes (**Figure 2-figure supplement 4B** [↗](#)). Further research is therefore required to identify the sequences and the E3 enzymes and deubiquitinating enzymes that control the turnover of this important protein class.

The highly dynamic landscape of phosphorylation during meiotic maturation

We performed phosphoproteomic mass spectrometry analysis in order to measure the overall dynamics of protein phosphorylation taking place during meiotic divisions and to identify which specific proteins and which sites are phosphorylated during this process. We identified 6783 different phospho-peptides and quantified the dynamics of relative change across all time points (see Appendix 5). These peptides came from 2308 distinct proteins. Our phosphosites dataset contains 80% Ser, 19.9% Thr, and 0.01% Tyr. Phospho-Tyr are slightly less abundant than what has been described in most cells (up to 0.05% (Sharma et al, 2014 [↗](#))). The same observation was made regarding the distribution of phosphorylated amino acids in mouse oocytes, where phospho-Tyr abundance is relatively diminished in oocytes compared to mouse organs (Sun et al, 2024 [↗](#)). When respective protein and un-phosphorylated peptide were also measured, we were able to compute the stoichiometry (see Methods). A list of all 415 phosphosites for which it was possible to calculate the phospho-occupancy is provided (see Appendix 6).

We identified the human homologous counterparts of 5901 (87%) among the 6783 sites found to be phosphorylated during meiotic divisions. Interestingly, only 177 (3%) out of 5901 conserved sites were already known to be phosphorylated on *phosphosite.org* [database \(https://www.phosphosite.org/homeAction\)](https://www.phosphosite.org/homeAction). Thus, our phosphosite dataset identifies a very large number of novel phosphosites, never previously identified and characterized. Such a small number of previously reported phosphorylations among those detected by our phosphoproteome might suggest that a large portion of these phosphorylation events are related to cell division. Dividing cells generally account for a small fraction of tissue mass. This leads to a high dilution of the phosphorylation signature of cell division, as compared to the highly synchronous meiotic divisions of oocyte maturation. Interestingly, 60% of the phosphosites detected in oocytes are dynamically regulated during meiotic maturation, highlighting the importance of this post-translational regulation in controlling this last step of oogenesis and meiosis. Unbiased clustering reveals that changes in protein phosphorylation are pervasive and far more dynamic than changes in protein abundance (see Appendix 3), which was also noted during yeast meiosis (Koch et al, 2024). Comparing the three stages of meiotic division (PRO, MI and MII) and based on the main cellular events of these stages, we can group phosphor-peptide stoichiometry into 5 classes (**Figure 3A**; see Appendix 5). Class I includes proteins bearing a phosphosite whose occupancy decreases during the first 2 hours following progesterone stimulation. This class includes only 136 phospho-peptides (2%) which might be involved in the early signaling pathway induced by progesterone and PKA downregulation. This small number was expected since there is a low level of phosphorylation detectable in prophase-arrested oocytes (Maller et al, 1977). This makes sense since there are presumably only a few substrates of PKA sufficient to initiate meiosis resumption. Class II is larger (39% of the phosphor-peptides) and includes proteins whose phosphorylation increases between NEBD and MI and which remain highly phosphorylated in both MI and MII. The massive amount of Class II phosphosites presumably reflects the catalytic power of Cdk1 as well as its downstream kinases, to generate the thousands of substrates essential for cell division. We used the online server (<https://kinase-library.phosphosite.org/kinase-library/score-site>) to score the phosphosites in the Class II and predict the kinases likely responsible for their phosphorylation (Johnson et al, 2023). Using a percentile score threshold of 90, we identified 303, 304, and 267 peptides predicted to be phosphorylated by Cdk1, Erk1/2, and Plk1, respectively (**Figure 3—figure supplement 1A**). Among these, 166 sites were predicted to be efficiently phosphorylated by either Cdk1 or Erk1/2, consistent with the similarity in their phosphorylation consensus motifs. In contrast, there was minimal overlap between the sites predicted to be phosphorylated by Plk1 and those targeted by Cdk1 or Erk1/2. Consequently, Class II phosphorylations are largely dependent on Cdk1 activity, whether they are directly catalyzed by this kinase or indirectly through kinases under its control, such as Erk1/2 and Plk1 (see below). Our measurements that most of the sites, which are phosphorylated in MI, remain stably phosphorylated during MI to MII transition is consistent with the observations made in yeast showing that most of Cdk1 motifs remain phosphorylated at the end of meiosis I (Celebic et al, 2024). Class III (5%) consists of proteins whose phosphorylation peaks in MI then decreases while oocytes progress throughout meiotic maturation. This is the most predominant class of phosphosites present during meiotic maturation in starfish (Swartz et al, 2021) and follow the peculiar pattern of Cdk1 activity peaking at MI in these oocytes (Okano-Uchida et al, 1998; Kishimoto, 2003). Class IV (9%) is composed of proteins whose phosphorylation progressively increases from MI to MII, hence potentially implicated in MII entry. Finally, Class V (5%) comprises proteins whose phosphorylation increases from MII and which could play a role in establishing the MII arrest. The phosphosites that are specific for either MI (Class III) or MII (Classes IV and V) are much less abundant (5%, 9% and 5% respectively, **Figure 3A**) and might distinguish the biochemical state of the two meiotic divisions. These conclusions were supported by the identification of the kinases whose predicted phosphorylation sites are most enriched within each class, by using the same server (<https://kinase-library.phosphosite.org/kinase-library/score-site>). Each class was associated with a distinct set of enriched kinases (**Figure 3-figure supplement 1B**). It confirms that Cdk1 and its downstream kinases, Erk1/2 and Plk1 are the master regulators in Class II. The kinases identified in Classes IV and V may represent novel regulators of entry into and arrest at MII.

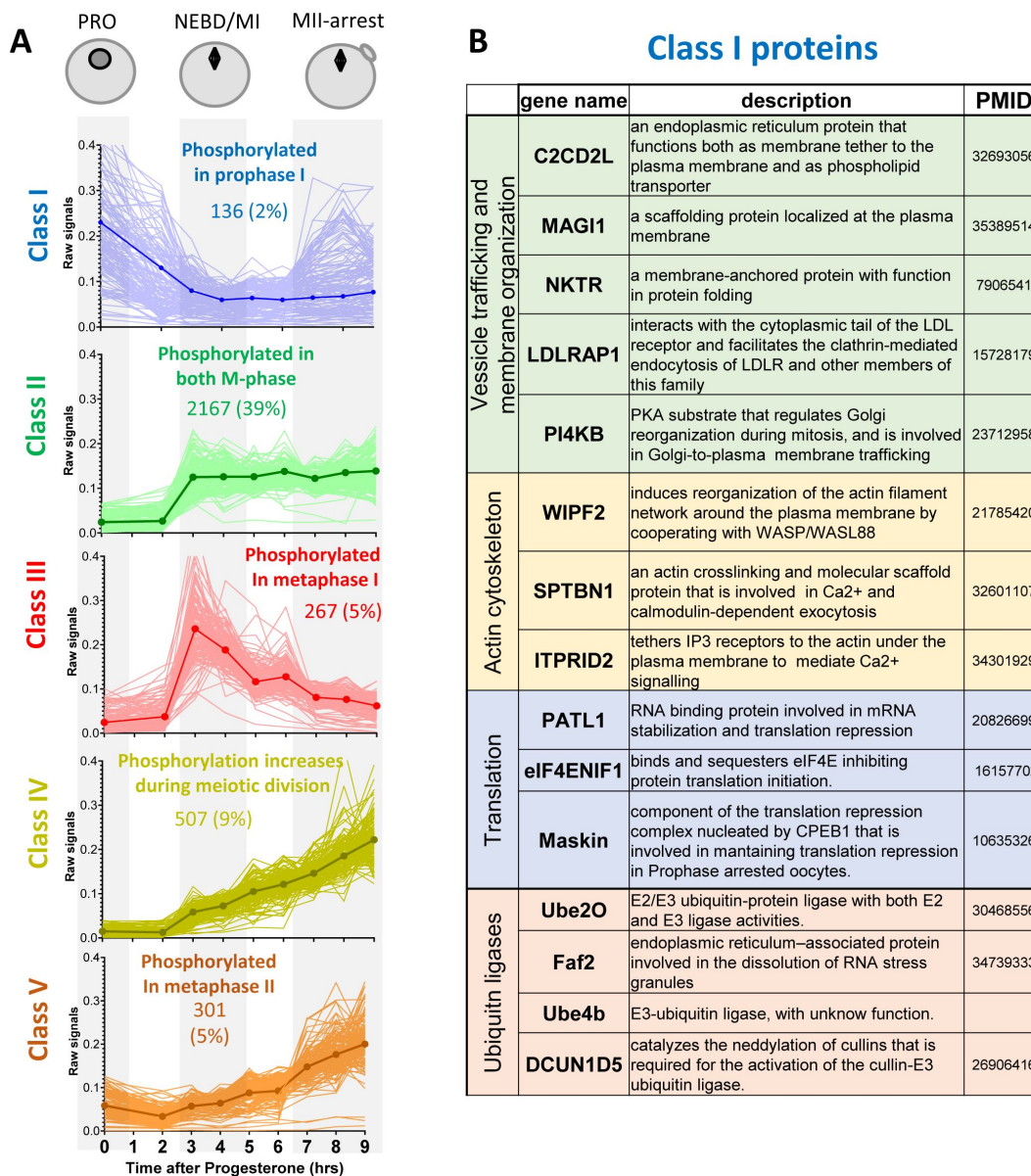


Figure 3.

Changes in protein phosphorylation during meiotic divisions.

(A) Relative phosphopeptide signals were normalized for the changes in the total level of the proteins. The changes in the phosphopeptides were measured between the three biological stages of meiotic divisions (PRO = 0hr, NEBD/MI = average of 3 and 4 hours, and MII-arrest = average of 7 to 9 hours). Class I (n=136 - 2%, in blue) includes proteins bearing a phosphosite whose occupancy decreases during meiosis resumption ($\text{Log2FC(MI/PRO)} < -1$). Class II (n=2167 - 39%, in green) contains proteins that are highly phosphorylated from MI ($\text{Log2FC(MI/PRO)} > 1$) to MII ($-1 < \text{Log2FC(MII/MI)} < 1$). Class III (n=267 - 5%, in red) consists of proteins whose phosphorylation is specific of MI ($\text{Log2FC(MI/PRO)} > 1$, $\text{Log2FC(MII/MI)} < -1$). Class IV (n=507 - 9%, yellow) is composed of proteins whose phosphorylation increases from MI ($\text{Log2FC(MI/PRO)} > 1$) to MII, picking in MII ($\text{Log2FC(MII/MI)} > 1$). Class V (n=301 - 5%, orange) comprises proteins whose phosphorylation does not vary in MI ($-1 < \text{Log2FC(MI/PRO)} < 1$) but are phosphorylated only in MII ($\text{Log2FC(MII/MI)} > 1$). The average signal of each Class was plotted with a thick and dark line. (B) Table summarizing the proteins bearing phosphosites belonging to Class I.

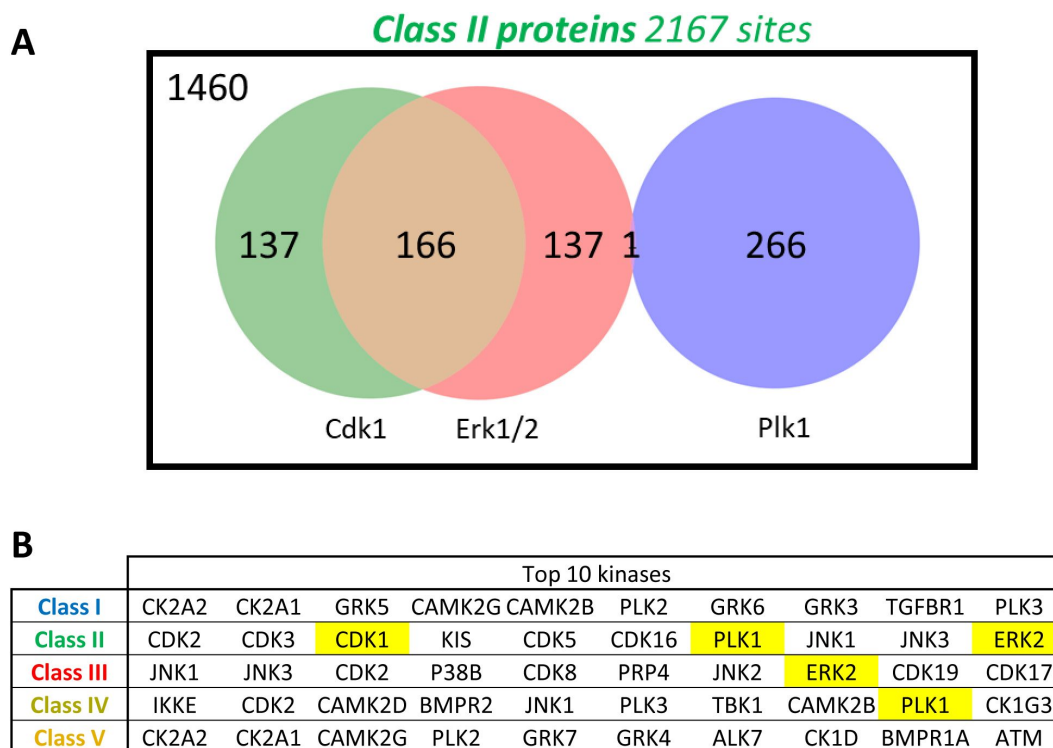


Figure 3-figure supplement 1.

(A) Venn diagram showing the number of Class II phosphosites predicted to be targeted by Cdk1, Erk1/2, and Plk1. Predictions were made using the kinase scoring tool available at <https://kinase-library.phosphosite.org/kinase-library/score-site>, applying a 90th percentile threshold. **(B)** Top 10 kinases whose predicted phosphorylation sites are most enriched in each class (I to V), using the kinase scoring tool available at <https://kinase-library.phosphosite.org/kinase-library/score-site>.

The release of the prophase arrest: receptors and membrane trafficking

Receptors

Since meiotic maturation signaling begins with the interaction of progesterone with membrane receptors, we first focused our analysis on these proteins (Josefsberg Ben-Yehoshua et al, 2007 [↗](#); Sadler & Maller, 1985 [↗](#)). Progesterone signals through activation of a seven-pass-transmembrane progesterone receptor (mPR β or PAQR8) that belongs to the progestin and adiponectin receptor family (Josefsberg Ben-Yehoshua et al, 2007 [↗](#); Nader et al, 2018 [↗](#), 2020 [↗](#)) (**Figure 1** [↗](#)). The release of the prophase block also involves the inactivation of an orphan constitutively active GPCR, GPR185, responsible for maintaining oocyte prophase arrest by ensuring high cAMP levels and PKA activity (Ríos-Cardona et al, 2008 [↗](#); Deng et al, 2008 [↗](#)). Progesterone decreases GPR185 signaling either through its cleavage by a metalloproteinase (Deng et al, 2008 [↗](#)) or through its endocytosis (Nader et al, 2014 [↗](#)). One limitation of our dataset lies in the loss of some membrane proteins during the fractionation protocol, explaining why our proteomic workflow did not detect mPR β and GPR185 or its close relative GPR12. Nevertheless, it does detect some proteins that interact with mPR, *i.e.* PGRMC1, VLDLR and APPL1. PGRMC1 is known to form a receptor complex with mPR α and is necessary for mediating progesterone signaling in zebrafish oocytes (Wu et al, 2018 [↗](#)). Interestingly, our phosphorylation data reveals that PGRMC1 and its close relative PGRMC2 are phosphorylated on two homologous threonine residues during meiotic maturation, suggesting that these proteins may contribute to mPR β activity in *Xenopus* oocytes. In the prophase-arrested oocyte, VLDLR regulates mPR trafficking from the endoplasmic reticulum through the Golgi to the plasma membrane, preparing the oocyte to be responsive to progesterone (Nader et al, 2018 [↗](#)). In contrast, progesterone induces clathrin-dependent endocytosis of mPR β into signaling endosomes, where mPR interacts transiently with APPL1 and Akt2 to induce meiosis (Nader et al, 2020 [↗](#)). The phosphoproteomic data reveal that both APPL1 and Akt are phosphorylated in response to progesterone but at the time of Cdk1 activation and on residues distinct from those identified by Nader et al. (2020) [↗](#). Hence, phosphorylation events may continue to regulate the trafficking and activity of this receptor during meiotic maturation, ensuring that the plasma membrane of the future egg is devoid of any steroid receptor.

Membrane traffic

Vesicular trafficking at the cell membrane appears to be crucial for the maintenance and the release the prophase meiotic arrest, and may also function through early embryogenesis (**Figure 1** [↗](#)). In the prophase-arrested oocyte, the accumulation of the GPR185 receptor at the plasma membrane maintains high cAMP levels (Nader et al, 2014 [↗](#)) while the plasma membrane targeting of mPR β renders the oocyte competent to respond to progesterone (Nader et al, 2018 [↗](#)).

Exocytosis is also crucial for the formation of the fluid-filled blastocoele cavity during embryogenesis (Müller, 2001 [↗](#)). Indeed, the apical membrane of the epithelium surrounding the blastula, where the polarized activities of ion channels and transporters generate the blastocoele fluid, is formed from the oocyte cell membrane (Müller, 2001 [↗](#)). This requires further remodeling. Progesterone rapidly blocks this exocytosis pathway (Colman et al, 1985 [↗](#); Leaf et al, 1990 [↗](#)), leading to a decrease in membrane surface area, which is revealed by the disappearance of microvilli that are enriched in oocytes but almost absent in eggs (Dumont, 1972 [↗](#); Kado et al, 1981 [↗](#); Larabell & Chandler, 1989 [↗](#); Bluemink et al, 1983 [↗](#)).

Importantly, blocking exocytosis induces meiotic maturation, in the absence of hormonal stimulation (Mulner-Lorillon et al, 1995 [↗](#); El-Jouni et al, 2007 [↗](#)). Furthermore, progesterone stimulates endocytosis of membrane proteins, such as GPR185, whose activity is suppressed by its

internalization, and mPR β , whose internalization is required to transduce its effects (Nader et al, 2018 [↗](#), 2020 [↗](#), 2014 [↗](#)). The stimulation of endocytosis also converts plasma membrane into intracellular vesicles that can provide membrane reserves, necessary to support the rapid cell divisions, and hence the requirement for greater total membrane surface area, during embryogenesis (Angres et al, 1991 [↗](#); Gawantka et al, 1992 [↗](#)). However, it is unknown whether PKA is responsible for the active secretory transport during the prophase arrest and how progesterone blocks exocytosis and stimulates endocytosis (**Figure 1** [↗](#)).

To improve our understanding of vesicular trafficking and membrane organization, we analyzed the proteins involved in these processes whose phosphosites belong to Class I (**Figure 3B** [↗](#)). Among these proteins, PI4KB deserves special attention. This protein is critical for the maintenance of the Golgi and trans-Golgi phosphatidylinositol-4-phosphate (PI4) pools. PI4KB regulates Golgi disintegration/reorganization during mitosis and is involved in Golgi-to-plasma membrane trafficking (De Matteis et al, 2013 [↗](#)). The phosphoproteome reveals that PI4KB is dephosphorylated within 2 hours after progesterone stimulation, on a PKA consensus phosphorylation site conserved in vertebrates. The phosphorylation of this site by PKA is known to control the interaction of PI4KB with Armadillo-like helical domain-containing protein 3 (ARMH3), an interaction important for the Golgi membrane integrity (McPhail et al, 2020 [↗](#); Isobe et al, 2017 [↗](#); Blomen et al, 2015 [↗](#)). PI4KB is also regulated by its interaction with ACBD3, an AKAP-like scaffold platform in Golgi, which directly binds the R regulatory subunit of PKA and regulates the traffic between Golgi and endoplasmic reticulum in a PKA-depending manner (Jia et al, 2023 [↗](#); Klima et al, 2016 [↗](#); Sasaki et al, 2012 [↗](#)). The phosphoproteome reveals that ACBD3 is phosphorylated during meiotic maturation at the time of NEBD (Class II). PI4KB is therefore at the crossroad between PKA and intra-membrane endoplasmic reticulum-Golgi-plasma membrane trafficking. Since these are key events involved in meiosis resumption, PI4KB is a particularly attractive candidate as a PKA substrate for future functional studies of membrane relocation.

Proteins involved in the actin cytoskeleton are also represented in Class I phosphosites (**Figure 3B** [↗](#)). Their early post-translational modifications could regulate the reorganization of the actin cytoskeleton that is known to accompany the modification of the secretory/endocytosis pathways (Gard, 1999 [↗](#)). Among them, ITPRID2/KRAP tethers IP $_3$ receptors, which are located in the membrane of the endoplasmic reticulum, to the actin under the plasma membrane, to mediate Ca $^{2+}$ signalling (Thillaiappan et al, 2021 [↗](#)). During meiotic maturation, the endoplasmic reticulum, which is the major Ca $^{2+}$ store, is enriched in the cortex of the oocyte (Campanella et al, 1984 [↗](#); Charbonneau & Grey, 1984 [↗](#); Terasaki et al, 2001 [↗](#)). This remodeling brings the Ca $^{2+}$ source close to its primary targets at fertilization, when Ca $^{2+}$ release from endoplasmic reticulum activates the oocyte and blocks polyspermy. The proteins highlighted above (**Figure 3B** [↗](#)) could participate to this critical reorganization of the endoplasmic reticulum.

Protein translation and accumulation orchestrate meiotic divisions

Two waves of translation take place in the oocytes: the first depends on PKA inactivation. It occurs upstream and independently of Cdk1 activity and is required for Cdk1 activation. The second translation wave takes place downstream Cdk1 activation (Santoni et al, 2024 [↗](#)). Hence, in contrast to mitosis, a period when protein synthesis is repressed (Ross, 1997 [↗](#)), translation is activated during meiotic maturation, indicating the existence of meiosis-specific controls of the process. However, the regulation of the two waves of protein translation and the identity of the newly translated proteins had not been unraveled. Interestingly, protein translation components are extensively regulated at the level of phosphorylation as demonstrated by the enrichment of the proteins regulating these processes among 3 of the 5 phospho - peptide classes (**Figure 4A** [↗](#)).

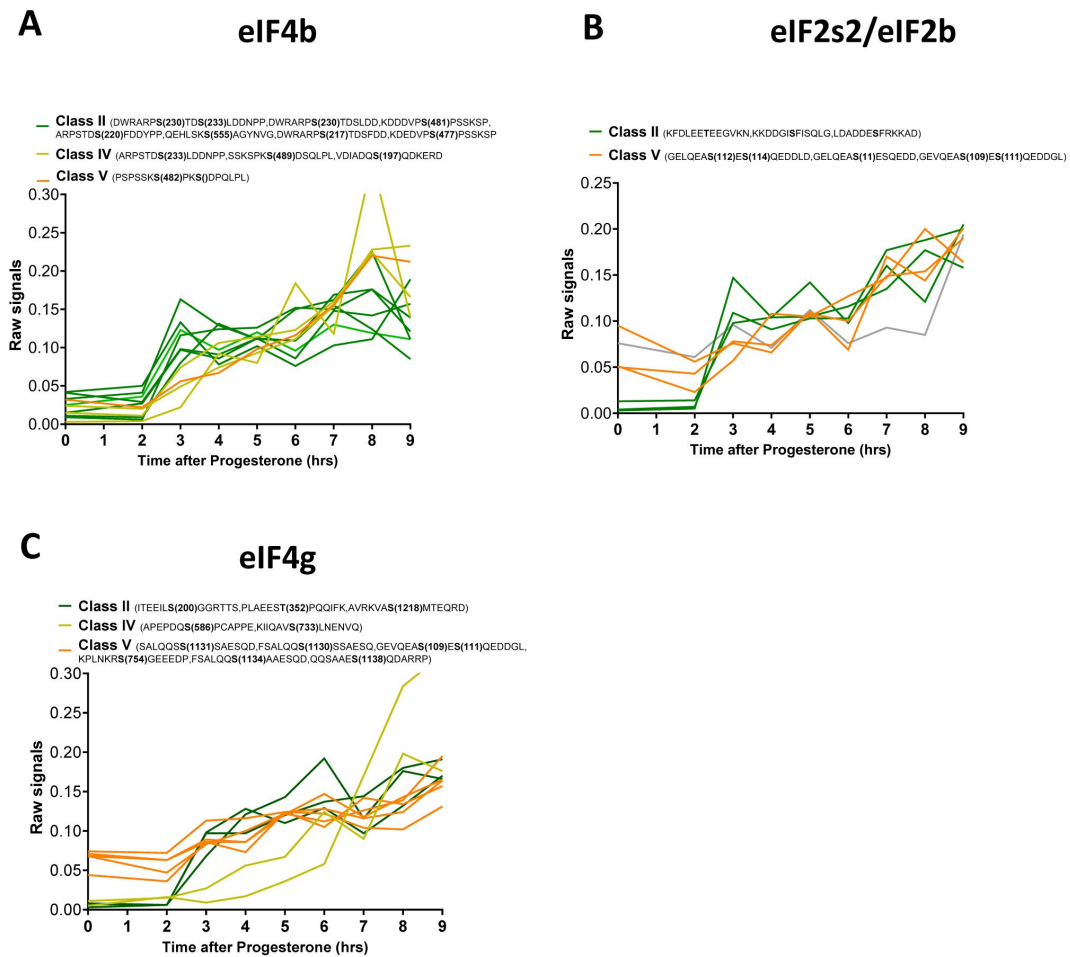


Figure 4-figure supplement 1.

Phosphorylation patterns of components of the initiation translation machinery during meiotic divisions.

(A-C) Levels of phosphorylation sites of eIF4b (A), eIF2s2/eIF2b(B) and eIF4g (C) during meiotic divisions are plotted. The phosphorylation curves are color-coded: Class II, green; Class IV, yellow, Class V, orange.

RBPs and translational control

Interestingly, several proteins implicated in the control of translation bear phosphosites belonging to Class I, as eIF4ENIF1/4E-T, PATL1 and TACC3/Maskin (**Figures 3B** and **4B**). eIF4ENIF1/4E-T is required for eIF4e1b localization to P-bodies where mRNAs are stored in a dormant state in zebrafish oocytes (Lorenzo-Orts et al, 2023). Both 4E-T and eIF4e1b are extensively phosphorylated on multiple residues during meiotic divisions, suggesting a highly dynamic regulation of P-bodies during this process (**Figure 4B**). Many RBPs known to control translation in oocytes are also regulated at the phosphorylation level, including Pum proteins (Nakahata et al, 2003; Ota et al, 2011; Padmanabhan, 2006), Zar proteins (Charlesworth et al, 2012; Yamamoto et al, 2013; Rong et al, 2019; Heim et al, 2022) and PATL proteins (Christou-Kent et al, 2018; Zhang et al, 2023; Marnef et al, 2010). PATL1 is a RNA-binding protein required for cytoplasmic mRNA P-body assembly in oocytes (Marnef et al, 2010). PATL1 and PATL2 have mutually exclusive expression patterns in *Xenopus* oocytes: PATL2 is degraded, as confirmed by our proteomic data (**Figure 2B,D**, **Figure 2-figure supplement 3A-B**), while PATL1 accumulates during meiotic maturation (Marnef et al, 2010) or is stable (**Figure 2B**, **Figure 2-figure supplement 3B**). Although the role of PATL2 and the regulation of its stability by phosphorylation have been well documented in the oocyte (Christou-Kent et al, 2018; Zhang et al, 2023), little is known about the role of PATL1 during meiosis. Both PATL1 and 2 inhibit translation when tethered to mRNA and assemble a complex that includes CPEB1, Xp54/DDX6, Rap55B/LSM14b (Marnef et al, 2010) (**Figure 4B**).

The early dephosphorylation of PATL1 detected in the phosphoproteome could modulate this inhibitory complex during early events of meiotic maturation, while the extensive phosphorylation of the other components of the complex could control its activity later during meiotic divisions (**Figure 4B**). Another component of the repressive complex is TACC3/Maskin. It interacts with CPEB1, although this interaction, and thus its role in regulation of translation, could not be reproduced in other studies (Minshall et al, 2007; Duran-Arqué et al, 2022). Importantly, besides its function in translation (Barnard et al, 2005; Groisman et al, 2002), TACC3/Maskin also regulates the centrosome-mediated microtubule nucleation through γ -TuRC (Peset et al, 2005). TACC3/Maskin is found widely regulated during meiotic divisions in the phosphoproteome (**Figure 4B**). Eight residues in TACC3/Maskin, of which only two are conserved in human Maskin (214/228 and 298/317 in human/*Xenopus*), exhibit a complex profile of phosphorylation (**Figure 4B**). Two sites identified in the phosphoproteome, S266 and S298, were previously found to be phosphorylated in egg extracts, regulating the function of TACC3/Maskin in the control of translation (Barnard et al, 2005). S266 was also reported to be phosphorylated in prophase oocytes and during oocyte meiotic maturation, although there is some controversy about the kinase (either PKA or Aurora-A) and the role of this phosphorylation in either the control of translation or centrosome attachment (Barnard et al, 2005; Pascreau et al, 2005). Interestingly, in our studies S240 is phosphorylated in the prophase oocyte and dephosphorylated early in response to progesterone (Class I) (**Figure 4B**). Since the sequence surrounding S240 does not correspond to a PKA consensus motif, it is hard to establish the biological significance; nevertheless, such an early dephosphorylation deserves further attention.

The control of translation initiation

Translation regulation might not only depend on the expression and post-translational regulation of the RBPs but could also involve the regulation of the core translation initiation machinery, since we find that many of its components are phosphorylated (**Figure 4C**). Indeed, eIF4b, an activator of the eIF4a helicase that facilitates ribosome scanning (Jackson et al, 2010), eIF2b/eIF2s2, a component of the molecular machinery recruitment on the met-tRNA (Jackson et al, 2010), and eIF4g, involved in the recognition of the mRNA 5'Cap (Jackson et al, 2010), are among the proteins bearing sites whose phosphorylation peaks in MII (**Figure 4-figure**

supplement 1 [↗](#)). These phosphorylation patterns correlate with the second wave of activation of translation that begins under the control of Cdk1 activity in MI and continues progressively through MII (Meneau et al, 2020 [↗](#); Santoni et al, 2024 [↗](#)).

The control of ubiquitin ligases

The levels of specific proteins are also regulated by proteolysis at each step of meiotic maturation. Targeted regulation allows for the accumulation of key proteins independently of any general increase in their translation, such as for Cyclin B1 (Santoni et al, 2024 [↗](#)), or the well-studied degradation of proteins, such as Cyclin B2, securin or CPEB1 (Taieb et al, 2001 [↗](#); Wassmann, 2022 [↗](#); Setoyama et al, 2007 [↗](#)). Interestingly, proteins controlling polyubiquitination display phosphor-peptides enriched in Class II, strengthening the view that a major regulation of protein turnover takes place at NEBD in order to orchestrate meiotic divisions (**Figures 3** [↗](#) and **5A** [↗](#)). APC is the key E3 ubiquitin ligase that promotes metaphase-anaphase transition. Although the phosphorylation status of some substrates determines their recognition by APC and influences the precise timing of their degradation, APC activity is also regulated by the phosphorylation of its core subunits (Bansal & Tiwari, 2019 [↗](#)) (**Figure 5B** [↗](#)).

Another important ubiquitination system that regulates stability of proteins during mitosis and meiosis is the SCF complex (Kinterová et al, 2022 [↗](#)). SCF is composed of three main components: Cul1, a scaffold protein; Skp1, which interacts with specific F-box proteins that are involved in the substrate recognition; and Rbx1, which interacts with the E2-ubiquitin ligases (Kinterová et al, 2022 [↗](#)) (**Figure 5C** [↗](#)). SCF ligases play important roles during oocyte meiotic maturation, especially the SCF^{βTrCP} complex. Indeed, SCF^{βTrCP} mediates the degradation of multiple cell cycle regulators as Emi1, Emi2/Erp1, CPEB1 and BTG4, whose degradation is required for progression through MI and MII in mouse and *Xenopus* oocytes (Kinterová et al, 2022 [↗](#)). Similar functions played by such protein degradations occur in mitosis, as Emi1 degradation that releases APC inhibition, then inducing Cyclin B destruction and mitotic exit (Reimann et al, 2001 [↗](#)).

Many F-Box proteins are also expressed in *Xenopus* oocytes, but they do not display high phosphorylation dynamics. An exception is Lmo7/Fbxo20, which is degraded during meiosis, and is dynamically phosphorylated (**Figure 5C** [↗](#)). Interestingly, Lmo7 overexpression causes defects at the spindle assembly checkpoint, affecting the progression through mitotic divisions (Tzeng et al, 2018 [↗](#)).

Additionally, many E2 enzymes were identified in our datasets as highly regulated at the translational/accumulation and post-translational levels during meiotic maturation (**Figure 5C** [↗](#)). This layer of regulation of protein ubiquitination remains understudied. Interestingly, an E2/E3 hybrid ubiquitin- protein ligase, UBE2O, UBE4B (also known as UBOX3 or E4) and FAF2 (or Ubxn3B/UBXD8) display phosphorylation sites belonging to Class I (**Figures 3B** [↗](#) and **5C** [↗](#)). FAF2 is involved in stress granule clearance (Gwon et al, 2021 [↗](#)). UBE2O is also known to be involved in the drastic proteome remodeling during erythroid differentiation (Nguyen et al, 2017 [↗](#)), as well as quality control factor for orphans of multiprotein complexes (Yanagitani et al, 2017 [↗](#)). Ribosomal proteins are known substrates of UBE2O which leads to their degradation (Nguyen et al, 2017 [↗](#)). Whether UBE2O is involved in the downregulation of ribosome biogenesis that occurs during meiotic maturation (Hyman & Wormington, 1988 [↗](#)), and in the decrease of Rpl26 (**Figure 2-figure supplement 4A** [↗](#)), is an interesting question. The dephosphorylation of UBE2O, UBE4B and FAF2 could also be involved in the accumulation of Cyclin B1, which was recently shown to occur independently of any increase in translation but to result from protein stabilization (Santoni et al, 2024 [↗](#)).

Another important regulator of the SCF system, DCUN1D5, is found regulated in our phosphoproteome. DCUN1D5 catalyzes the neddylation of all cullins, which is necessary for the activation of cullin-RING E3 ubiquitin ligases (Keuss et al, 2016 [↗](#)). DCUN1D5 is dephosphorylated

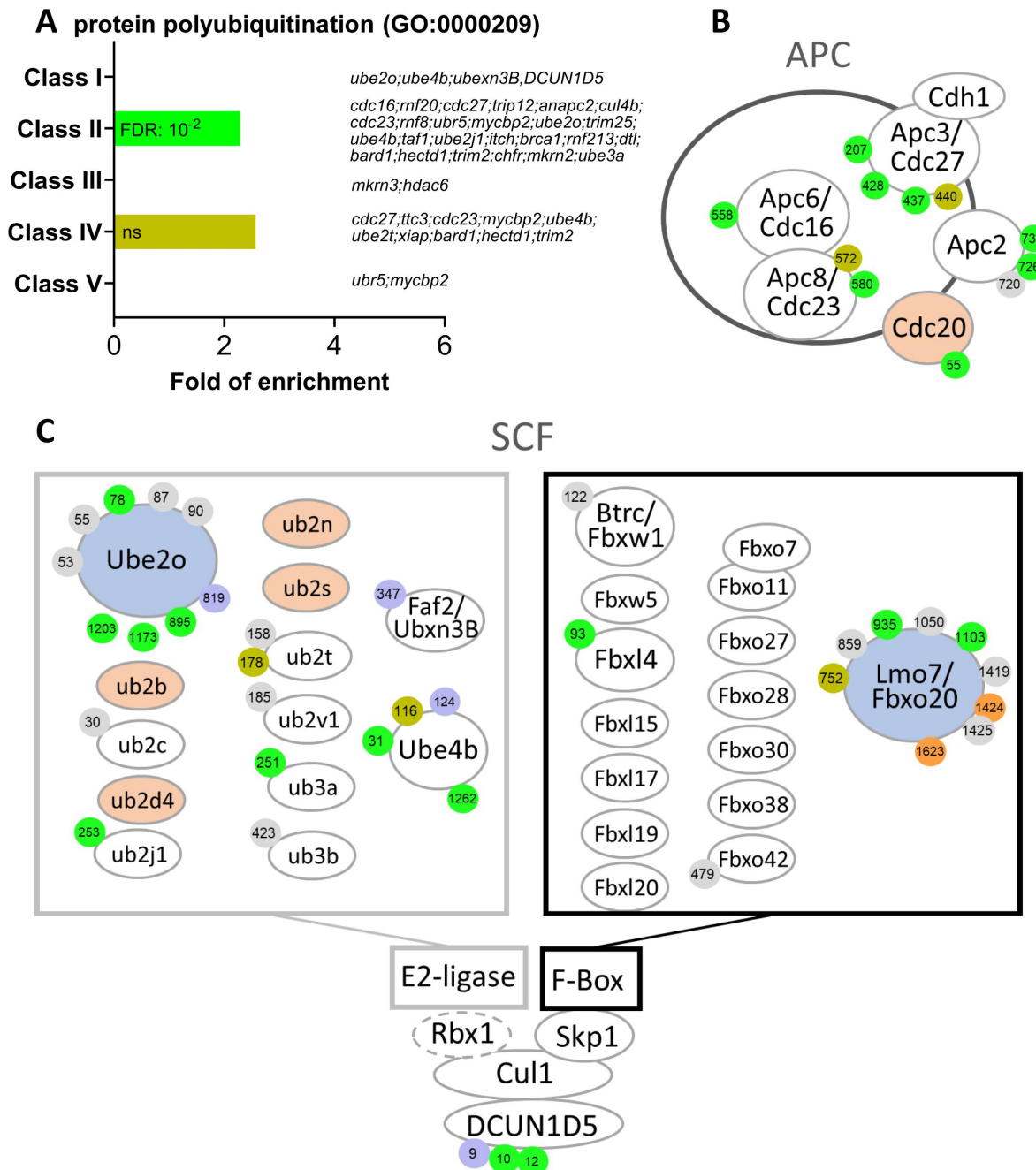


Figure 5.

E2 and E3 ubiquitin ligases are highly regulated by phosphorylation.

(A) Gene ontology enrichment analysis of the biological process “protein polyubiquitination” (GO:0000209) among the phosphopeptides found in each class defined in **Fig. 3A**. (B-C) Graphical representation of the phosphorylation dynamics of the components of two E3 ubiquitin ligases, the anaphase promoting complex (APC) (B) and the Skp, Cullin, F-box containing complex (SCF) (C) during meiotic divisions. Proteins are color-coded to highlight changes in protein abundance during meiotic divisions: unchanged=white, decreased=blue, increased=light orange. The phosphorylated sites are color-coded: unchanged, grey; Class I, blue; Class II, green; Class III, red; Class IV, yellow; Class V, orange. The numbering is referred to the iso-allele which has higher number of phosphorylation sites detected.

at S9, a PKA consensus site, and phosphorylated starting NEBD at S10 and S12 (**Figure 5C**). These phosphorylations could control DCUN1D5 ability to activate SCF, especially since the inhibition of neddylation causes a meiotic arrest in MI in mouse oocytes (Yang et al, 2019). Altogether, our dataset provides attractive candidates to be involved in the regulation of both translation and protein stability that occur during meiotic maturation.

Activation of Cdk1 occurs through an intricate network of phosphorylation of Cdk1 regulators

The activation of Cdk1 during oocyte maturation relies on an intricate network of feedforward and feedback phosphorylation pathways mediated by kinases and phosphatases (**Figure 1**). Indeed, components of the cell division machinery are highly enriched among Class II, III, IV and V (**Figure 6A**). Some of the phosphorylation sites implicated in cell division control have been identified previously through detailed studies using site-specific mutagenesis and phospho-specific antibodies. Our phosphoproteomic analysis of oocyte maturation provides new regulatory elements (**Figure 6B**). An important regulator of the Cdk1 auto-amplification loop is the kinase Plk1 that contributes to the phosphorylation of the two direct Cdk1 regulators, Cdc25 and Myt1. Plk1 is activated by Aurora-A, a kinase itself under the indirect control of Cdk1 in *Xenopus* oocyte (Maton et al, 2003), through the phosphorylation of T210 (T201 in *Xenopus*) (Macůrek et al, 2008), a residue localized in the activation T-loop of the kinase domain. We show here that Plk1 phosphorylation at T210 increases in MI and is then constant throughout meiotic maturation (Class II) (**Figure 6B** and EV2A). This is consistent with the function of Plk1, which phosphorylates multiple proteins required for the mechanics of division, and which must logically be kept active throughout both meiotic divisions (Hansen et al, 2006; Solc et al, 2015). Interestingly, the phosphoproteome also detects additional phosphorylation sites of Plk1, S326 (S335 in human) and S340 (not conserved in human), which were previously found to be upregulated in response to okadaic acid (Wind et al, 2002), an inhibitor of PP2A that strongly induces M-phase. These two residues are located between the two functional domains of Plk1, the kinase domain and the Polo-box domain (Cheng, 2003). This suggests that the phosphorylation of other critical residues than T210 of Plk1 could be essential for its catalytic activation. Aurora-A kinase is activated during meiotic maturation downstream Cdk1 (Maton et al, 2003; Castro et al, 2003; Maton et al, 2005). While our phosphoproteome does not detect Aurora-A autophosphorylation at T288 (T295 in *Xenopus*), it registers the extensive phosphorylation of Bora, an activator of Aurora-A involved in Plk1 activation (Thomas et al, 2016) (**Figure 6B-figure supplement 1B**). Several phosphorylation sites of Bora have been reported to be important for its function as an Aurora-A activator: S41/38, S112/110, S137/135 and S252/S252 (human/*Xenopus*) (Thomas et al, 2016; Vigneron et al, 2018; Tavernier et al, 2015, 2021). Among these sites, S110 phosphorylation of Bora by Cdk1-Cyclin A is critical for mitotic entry in *Xenopus* egg extracts (Vigneron et al, 2018; Tavernier et al, 2015, 2021). Our phosphoproteome reveals that Bora belongs to Class II, being phosphorylated starting NEBD, but surprisingly on distinct sites that were never reported in the literature: S191/189, T294/287, S552/549, S-/521, T12/9 and T15/12 (human/*Xenopus*), with the exception of S278/285. In contrast to the situation in mitosis, Cyclin A is very weakly expressed in the prophase oocyte and accumulates during meiotic maturation (Kobayashi et al, 1991). Our proteome dataset reveals that Cyclin A accumulates after MI. Hence, the phosphorylation of Bora at MI cannot be catalyzed by Cdk1-Cyclin A in the *Xenopus* oocyte. Cdk1-Cyclin B is probably involved in these phosphorylations that take place concomitantly with its activity and at Cdk1 phosphorylation consensus sites for 5 of the 7 phosphosites (**Figure 6B-figure supplement 1B**). Thus, this analysis reveals that Bora activation correlates with its phosphorylation at residues not previously identified in earlier studies of the oocyte. We also detect additional phosphorylation events on other players of the Cdk1 amplification loop, as Cdc25 and Gwl/MASTL (**Figure 6B-figure supplement 1C-D**), whose extensive phosphorylation during M-phase is usually detected by a large electrophoretic shift (Dupré et al, 2014).

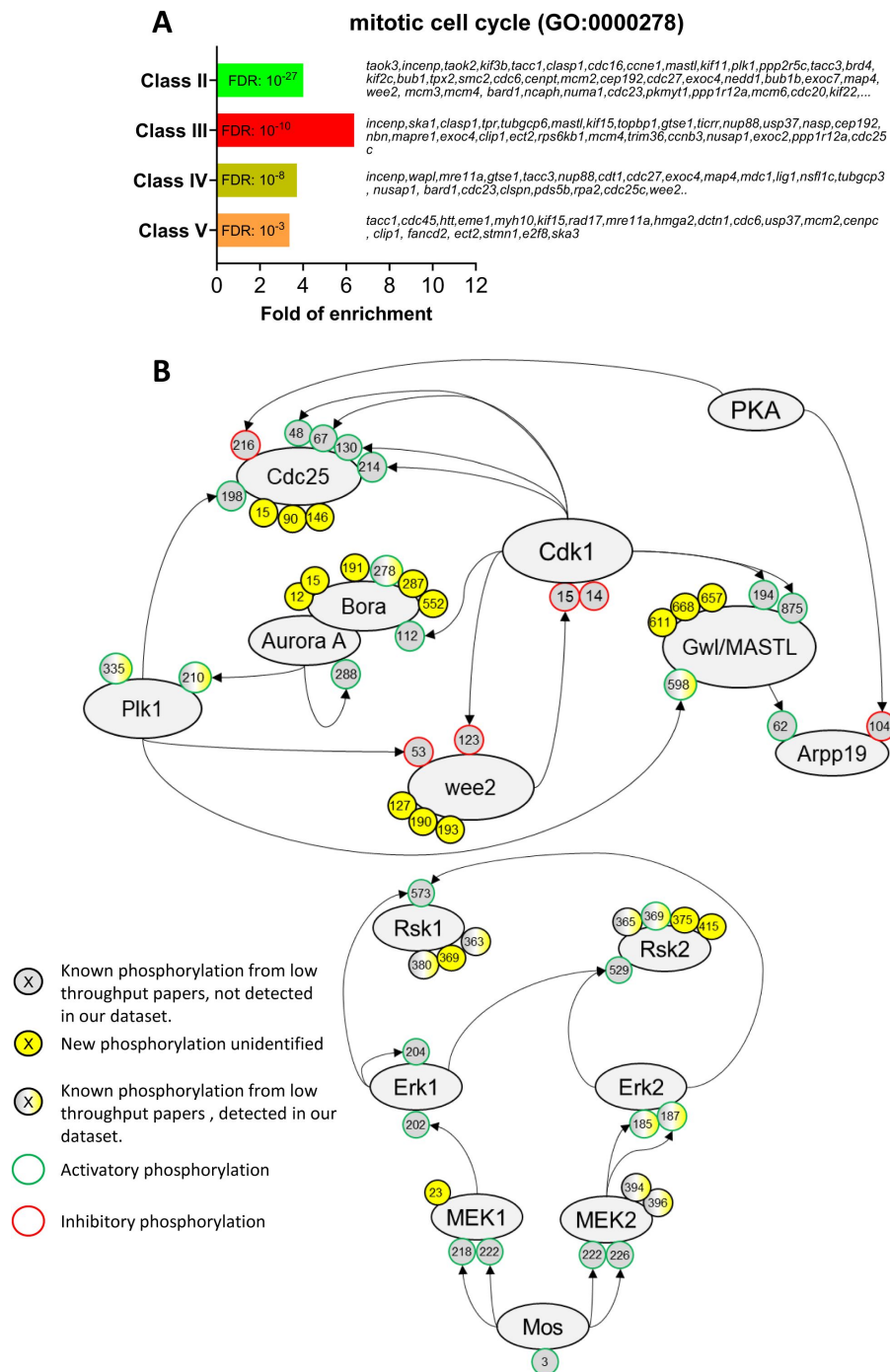


Figure 6.

The network of phosphorylations regulating Cdk1 and the Mos/MAPK pathway.

(A) Gene ontology enrichment analysis of the biological process “mitotic cell cycle” (GO:0000278) among the phosphopeptides found in each class defined in Fig. 3A. (B) Graphical representation of the network of phosphorylations of cell cycle regulators controlling Cdk1 activation (upper panel) and the Mos/MAPK pathway (lower panel). The human nomenclature numbering has been used for the phospho-sites. The phosphosites are displayed by dots and color-coded: yellow background, new sites identified in this paper and never described elsewhere; grey background, phosphosites already documented in vertebrates by low throughput papers and not identified in this paper; grey-yellow background, phosphosites already documented in vertebrates by low throughput papers and also identified in this paper. The functional effect of the phosphorylation is indicated by the color of the dot contour line: green, activatory; red, inhibitory; black, unknown.

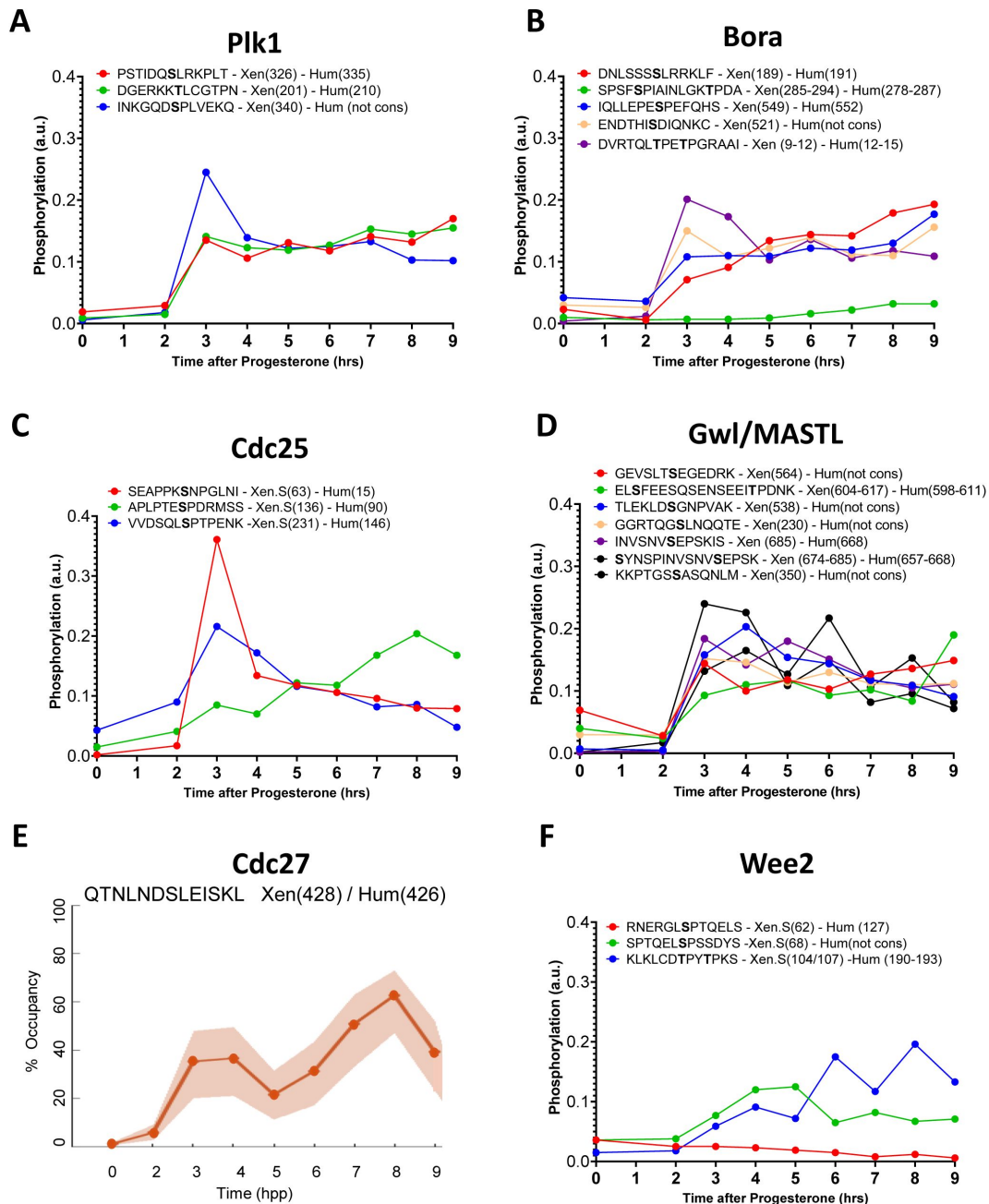


Figure 6-figure supplement 1.

Phosphorylation patterns of Cdk1 regulators during meiotic divisions.

(A-D and F) Levels of phosphorylation sites of Plk1 (A), Bora (B), Cdc25 (C), Gwl/MASTL (D) and Wee2 (F) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature. (E) The phosphorylation occupancy for S428 of Cdc27 is plotted with its confidence intervals.

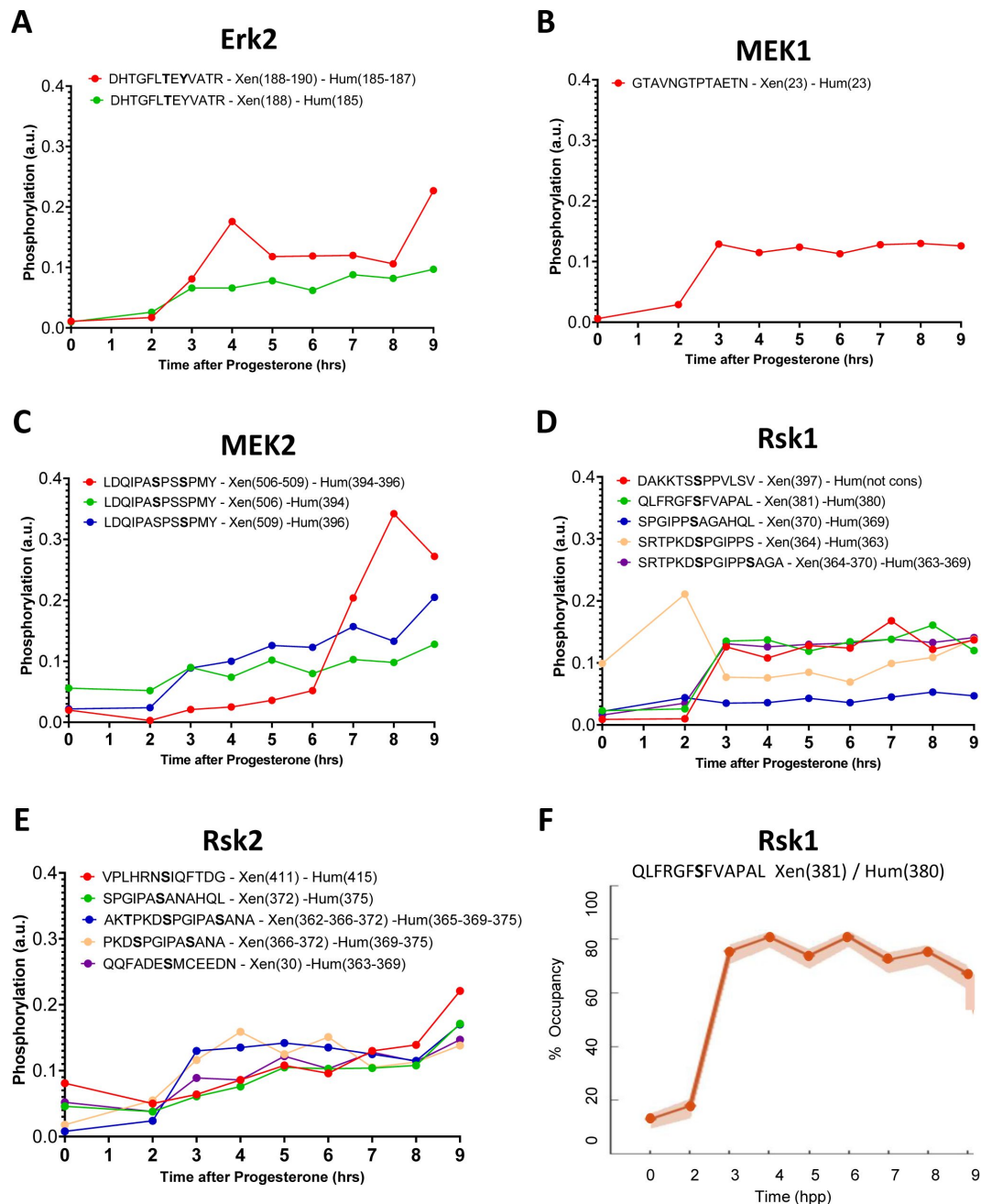


Figure 6-figure supplement 2.

Phosphorylation patterns of the mos/MAPK pathway during meiotic divisions.

(A-E) Levels of phosphorylation sites of Erk1 (A), Bora (B), MEK1 (C), MEK2 (D) and Rsk2 (E) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature. (F) The phosphorylation occupancy for S381 of Rsk1 is plotted with its confidence intervals.

Phosphoregulation of the bimodal activity of Cdk1 during MI-MII transition

In *Xenopus* oocytes, Cdk1 is not fully inactivated at anaphase I and during extrusion of the 1st polar body. Indeed, the low Cdk1 activity that persists during the MI-MII transition is essential to avoid the reformation of a nucleus, chromosome decondensation and DNA replication (Huchon et al, 1993 [\[1\]](#); Thibier et al, 1997 [\[2\]](#)). This short period is followed by a further increase of Cdk1 activity, due to a sustained synthesis of B-cyclins; the increased Cdk1 activity allows the formation of the MII spindle (Gerhart et al, 1984 [\[3\]](#)) (**Figure 1** [\[4\]](#)). Although this bimodal subtle regulation of Cdk1 activity is essential for the success of meiosis, *i.e.* linking two successive divisions without intervening DNA replication, it has never been quantified precisely. Cdk1 activity can be measured by looking at the phosphorylation of one of its direct substrates, Cdc27, a core APC component, at S428 (S426 in human) (Patra & Dunphy, 1998 [\[5\]](#)). Since PP2A, the phosphatase responsible for dephosphorylating Cdc27, is inactive during the MI-MII transition, the phosphorylation levels of Cdc27 can be directly attributed to Cdk1 activity during this period (Deák et al, 2003 [\[6\]](#); Torres et al, 2010 [\[7\]](#); Lemonnier et al, 2021 [\[8\]](#); Labbé et al, 2021 [\[9\]](#)). Our precise quantification of the phospho-occupancy of this site shows that its phosphorylation is virtually absent in prophase, increases by 40% in MI, decreases during the MI-MII transition to 20%, then increases again in MII by 60% (**Figure 6-figure supplement 1E** [\[10\]](#)). Three other residues of Cdc27, T205/207, S435/437 and S438/440 (human/*Xenopus*), also known to be phosphorylated at M-phase (Kraft, 2003 [\[11\]](#)), follow the same bimodal pattern than S428. The subsequent increase in Cdk1 activity essential for MII entry requires that the kinase escapes inhibitory phosphorylations by Myt1 and Wee2. Both Myt1 and Wee2 are reported to be inhibited by phosphorylations, although the sites responsible for this inhibition are not precisely defined. In *Xenopus*, Wee2 is not expressed in prophase oocytes and accumulates after NEBD (Nakajo et al, 2000 [\[12\]](#); Charlesworth et al, 2012 [\[13\]](#)) (**Figure 1** [\[4\]](#)). Our analysis reveals that newly synthesized Wee2 is immediately phosphorylated at 4 sites (Classes II and IV), one of them being not conserved in human, and none of them corresponding to the proposed inhibitory S53 and S123 sites (**Figure 6A-B-figure supplement 1F** [\[14\]](#)). Hence, these data provide exciting new avenues of research to discover unexpected new regulations of old players in Cdk1 activation.

The Mos/MAPK pathway activation

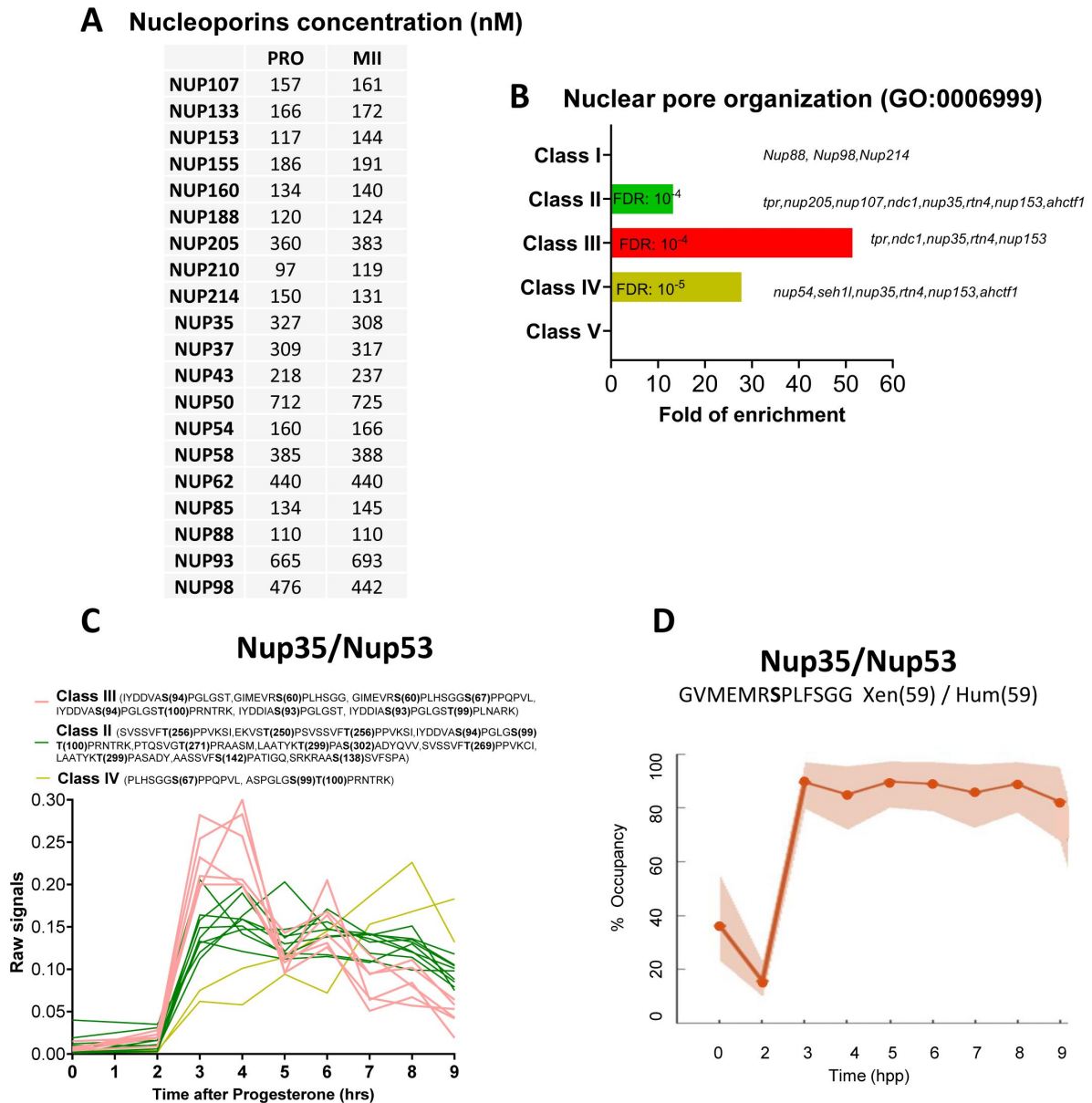
Mos plays a well-known role at the origin of a critical pathway controlling meiotic maturation (Yew et al, 1992 [\[15\]](#); Furuno et al, 1994 [\[16\]](#)). It phosphorylates Mek1/2, which in turn phosphorylates Erk1/2 (also known as MAPK) which leads to Rsk1/2 activation (**Figure 6B** [\[17\]](#)). These play multiple essential roles, such as preventing the reformation of a replicating nucleus between MI and MII (Dupre et al, 2002 [\[18\]](#)), enabling arrest in MII (Sagata et al, 1989b [\[19\]](#)), but also participating in Cdk1 activation (Sagata et al, 1989a [\[20\]](#)) and the functioning of meiotic spindles (Araki et al, 1996 [\[21\]](#); Choi et al, 1996 [\[22\]](#); Bodart et al, 2005 [\[23\]](#)). The phosphoproteome detects the activation of the Mos-MAPK module, with the phosphorylation of Erk2 on T185 and Y187 (T188 and Y190 in *Xenopus*) by MEK1/2 (Payne et al, 1991 [\[24\]](#)), an event that begins in MI (**Figure 6B-figure supplement 2A** [\[25\]](#)). Interestingly, additional phosphorylation sites are identified in MEK1 and Rsk1/2, the latter being hyperphosphorylated during meiosis as detected by a large electrophoretic shift (Bhatt & Ferrell, 2000 [\[26\]](#)) (**Figure 6B-figure supplement 2B-E** [\[27\]](#)). Rsk1 auto-phosphorylation at S381 (S380 in human) is required for its activity (Vik & Ryder, 1997 [\[28\]](#)). Our phosphoproteome measures a highly-confident phospho-occupancy for this site demonstrating the switch-like activation of the MAPK pathway at NEBD, followed by the maintenance of a constant level of its activity (**Figure 6B-figure supplement 2F** [\[29\]](#)). The timing of activation of the Mos-MAPK cascade has been controversial, one hypothesis being that it is activated before Cdk1 and participates in the activation of this kinase (Sagata et al, 1988 [\[30\]](#); Gavin et al, 1999 [\[31\]](#)), while another is that it is under the control of the starter Cdk1 activity and is only involved in the autoamplification loop (Santoni et al, 2024 [\[32\]](#); Dupre et al, 2002 [\[18\]](#); Fisher et al, 1999 [\[33\]](#)). The phosphoproteomic data clearly support the latter.

Phosphorylation of key components of the nuclear envelope during breakdown

As a result of the massive activation of kinase activities under the control of Cdk1, hundreds of proteins are phosphorylated and orchestrate the structural events choreographing the mechanics of meiotic divisions. The first of these is the massive vesiculation of the nuclear envelope surrounding the enormous oocyte nucleus. During mitosis, nuclear pore proteins are extensively phosphorylated by Cdk1-Cyclin B, Plk1 and Nek6/7 to promote the breakdown of the nuclear envelope (Kutay et al, 2021 [\[1\]](#)). The *Xenopus* oocyte has the unusual feature of having an enormous nucleus (475 μm in diameter), and therefore a considerable surface area of the nuclear envelope (0.96 mm^2), as well as large stocks of annulate lamellae in the cytoplasm. Annulate lamellae are cytoplasmic stack cisternae of nuclear envelope perforated with nuclear pores, 10 times more abundant than in the nucleus in the *Xenopus* oocyte, but devoid of lamins (Bement & Capco, 1990 [\[2\]](#); Miller & Forbes, 2000 [\[3\]](#); Cordes et al, 1995 [\[4\]](#)). Their surface area also far exceeds that of the nuclear envelope surrounding the nucleus (Cordes et al, 1995 [\[4\]](#)). This correlates with the high protein concentrations of the nuclear pore proteins (0.1 to 0.7 μM) that we have measured (Figure 7A [\[5\]](#)). Annulate lamellae play a critical role as a reservoir for the formation of the 4000 of nuclei that form during the embryonic cell divisions leading up to the mid-blastula transition and the onset of transcription (Kessel et al, 1986 [\[6\]](#); De Magistris & Antonin, 2018 [\[7\]](#); Feldherr, 1974 [\[8\]](#)). During meiotic maturation, NEBD and vesiculation of annulate lamellae occur at the same time, in a progressive spatial manner, starting from the vegetal pole. Importantly, the nucleus and annulate lamellae do not re-form between the two meiotic divisions, reducing the risk that an S-phase can take place between MI and MII (Bement & Capco, 1990 [\[2\]](#)). A gene ontology analysis of our phosphoproteome highlights that the phosphorylation of nuclear pore components is enriched in Classes II, III and IV, displaying the highest level of enrichment in Class III, hence characteristic of MI (Figure 7B [\[5\]](#)). The nucleoporin Nup53 (also called Nup35) is a component of the soluble core region of the nuclear pore complex that is extensively phosphorylated in mitosis by Cdk1 and Plk1 (Linder et al, 2017 [\[9\]](#)). Phosphomimetic mutations of all Cdk1 and Plk1 sites in Nup53 slow down the reformation of nuclear pores in interphase and mutation to phospho-null amino acids impairs the nuclear breakdown during meiosis (Linder et al, 2017 [\[9\]](#)). Our phosphoproteome reveals that Nup53 phosphorylation is more dynamic than previously believed (Figure 7C [\[5\]](#)). Indeed, 6 phospho-peptides are found to peak in MI (Class III), 9 phospho-peptides are phosphorylated both in MI and MII (Class II), and 2 phospho-peptides continuously increase in phosphorylation during meiotic maturation (Class IV) (Figure 7C [\[5\]](#)). We were able to calculate the phospho-occupancy of the S59 (S66 in human), showing that it increases from 30% in prophase to 90% in MI oocytes and stays constant until MII (Class II) (Figure 7D [\[5\]](#)). Several other transmembrane nucleoporins, notably NDC1 and POM121C, undergo dramatic increase of phosphorylation at transition to MI. Altogether, these observations indicate that dissociation of the nuclear envelope and the annulate lamellae occurring in MI might require either a specific set of phosphorylations and/or a general level of phosphorylation that is higher than the level of phosphorylation required to maintain the nucleus and the annulate lamellae dissociated during the MI- MII transition through MII.

The regulation of centrosomal proteins during oocyte maturation

An intriguing feature of oocytes in most, if not all, metazoan species is the elimination of their centrioles. This allows the sperm to contribute its centriole and thereby initiates the division cycle and the cell cycle. There is longstanding literature that proves that the oocyte or egg has retained its ability to generate centrioles. However, in nature, the female centrioles are lost and the sperm centriole is the main contributor. In *Xenopus*, oocytes entering prophase I contain a typical centrosome, including a pair of centrioles and pericentriolar material (PCM) containing γ -tubulin and pericentrin. Centrioles disappear at the end of the pachytene stage of prophase I and PCM disperses in multiple foci in the cytoplasm that are no longer functional as microtubule nucleation sites (Gard et al, 1995 [\[10\]](#)). At fertilization, the male gamete delivers two centrioles but does not



provide the PCM (Delattre & Gönczy, 2004 [↗](#)). Therefore, in most species (with the exception of rodents), the complete centrosome of the one-cell stage embryo is a bi-parental inheritance, reconstituted from paternal centrioles and maternal PCM (Delattre & Gönczy, 2004 [↗](#)). In *Xenopus*, oocytes store enough quiescent centrosomal building blocks for duplicating and assembling 1000 to 2000 centrosomes (Gard et al, 1990 [↗](#)). This implies that oocytes should comprise an enormous reserve of centrosomal proteins, *i.e.* centriolar and PCM proteins, that sustain the reconstitution of a functional centrosome at fertilization and its subsequent rounds of duplication in the early embryo.

Centriolar proteins

The centriole is formed by a polarized cylinder of microtubule triplets, decorated along the proximal to distal axis with several appendages that define the three main regions of the organelle: the proximal part, the central core, and the distal region (LeGuennec et al, 2021 [↗](#)). In agreement with a published RNA-seq and proteome of *Xenopus* eggs (Session et al, 2016 [↗](#); Wühr et al, 2014 [↗](#)), our proteome reveals that most of the centriolar proteins are expressed in the oocyte (**Table 1** [↗](#)). The components of the proximal part, as SAS-6 and CPAP/CENPJ, and of the central core, as POC1B, POC5 and Centrin 1, are either constant or accumulate during meiotic maturation (**Table 1** [↗](#)). In contrast, proteins of distal and subdistal appendages are present at lower concentration, either constant or decreasing during meiotic maturation (**Table 1** [↗](#)). Indeed, the appendages are critical for the docking of mature centrioles at the plasma membrane and ciliogenesis (Tanos et al, 2013 [↗](#)), a process that is not functional in the oocyte and during the early embryonic divisions (Drysdale & Elinson, 1992 [↗](#)).

Besides the basic building blocks forming the centriole, the oocyte is expected to provide the machinery for its duplication, the first cycle occurring immediately after fertilization. The key regulators of centriole duplication, including Plk4, SAS-6 and STIL, are generally expressed at very low levels (Bauer et al, 2016 [↗](#)). While Plk4 is expressed at the mRNA level in oocytes but escaped our proteomic detection, SAS-6 and STIL are expressed at a constant level during meiotic maturation (**Table 1** [↗](#)). The estimated concentrations of both proteins are in agreement with the previous predictions, that cartwheels comprise one molecule of STIL for every SAS-6 dimer (Bauer et al, 2016 [↗](#)). Other important regulators of centriole duplication, such as CEP192, CEP152, CEP63, RTTN, CEP110/CCP110, CEP97 and CEP76 were also detected, and are either constant or accumulate during meiotic divisions (**Table 1** [↗](#)). Interestingly, our phosphoproteome reveals that several of these proteins are phosphorylated during meiotic maturation. STIL is phosphorylated at S554, which is located in a Cdk1 phosphorylation consensus (Appendix Figure S5A), an interesting observation since phosphorylation of STIL by Cdk1-Cyclin reduces the efficiency of daughter centriole assembly. This phosphorylation provides an explanation why centrioles cannot duplicate during M-phase (Zitouni et al, 2016 [↗](#); Steinacker et al, 2022 [↗](#)). A massive CEP97 phosphorylation occurs at MI at 8 distinct sites, only 3 of them lying within a Cdk1 consensus motif (Appendix Figure S5B). Remarkably, one of the phosphosites of CEP97, S633, has been recently shown to be phosphorylated by the kinase Dyrk1a in multiciliated cells of *Xenopus* embryos (Lee et al, 2022 [↗](#)); it is required for centriole duplication. Therefore, although centrioles are not assembled, all centriolar proteins are expressed in the full-grown oocyte and these components serve as a stock for the biogenesis of the centrioles of the embryo. Our results suggest that specific phosphorylations of proteins regulating centriole assembly could prevent the *de novo* establishment of centrioles in the oocyte.

PCM proteins

The core element of the PCM is the γ -tubulin ring complex (γ -TuRC) which is the basic element for nucleation of microtubules from microtubule-organizing centers (MTOCs) (Oakley & Oakley, 1989 [↗](#)). It includes γ -tubulin that is combined with other proteins known as GCP2/tubgcp2 to GCP6/tubgcp6. Some PCM proteins or regulators of PCM assembly have been described in functional studies to be present in the oocyte and then recruited to the zygotic centrosome as

	Gene name	Proteome										(Wuhr,2014)	(Session,2016)
		0	2	3	4	5	6	7	8	9	Log2 FC (MII/PRO)	Estimated concentration in MII-arrested oocytes (nM)	Count per Million (CPM) in PRO I
Proximal part	SAS-6	0.11	0.09	0.10	0.10	0.09	0.09	0.09	0.09	0.08	-0.3	36	20
	CENPJ	0.07	0.09	0.09	0.10	0.10	0.11	0.09	0.10	0.09	0.4	9	56
	Cep135	not detected										5	1
Central core	POC1B	0.05	0.05	0.07	0.08	0.09	0.14	0.11	0.10	0.11	1.2	7	96
	POC5	0.05	0.06	0.07	0.08	0.09	0.10	0.10	0.11	0.12	1.1	39	100
	CETN1	0.08	0.08	0.11	0.09	0.09	0.09	0.10	0.09	0.10	0.2	not detected	3
	SFI1	not detected										3	7
	POC1A	not detected										20	54
	CETN2	not detected										82	not detected
	WDR90	not detected										5	25
	FAM161A	not detected										not detected	34
Distal appendages	SCLT1	0.10	0.09	0.10	0.11	0.10	0.10	0.08	0.08	0.08	-0.4	13	2
	FBF1	0.11	0.12	0.11	0.12	0.11	0.11	0.08	0.07	0.06	-0.7	not detected	17
	CEP83	not detected										not detected	7
	CEP89	not detected										not detected	1
Sub-distal appendages	FBF1	0.11	0.12	0.11	0.12	0.11	0.11	0.08	0.07	0.06	-0.7	not detected	17
	CEP164	0.10	0.08	0.11	0.09	0.12	0.09	0.08	0.06	0.08	-0.4	1	not detected
	NIN	0.11	0.09	0.11	0.07	0.08	0.11	0.10	0.08	0.10	-0.2	4	27
	TUBE1	0.10	0.09	0.10	0.10	0.10	0.11	0.09	0.09	0.09	-0.1	7	8
	TUBD1	0.08	0.08	0.08	0.09	0.09	0.11	0.09	0.09	0.10	0.1	13	28
	ODF2	not detected										7	8
	CEP128	not detected										3	17
	CNTRL	not detected										6	26
	CEP170	not detected										4	3
	TCHP	not detected										2	16
	CCDC68	not detected										not detected	3
	tubz1	not detected										not detected	10
	CCDC120	not detected										not detected	not detected
Centriole duplication	CCP110	0.06	0.07	0.06	0.08	0.08	0.09	0.10	0.12	0.12	0.9	7	20
	CEP152	0.07	0.07	0.10	0.10	0.10	0.12	0.10	0.11	0.09	0.6	7	161
	CEP192	0.08	0.08	0.08	0.08	0.09	0.09	0.10	0.11	0.10	0.5	21	24
	STIL	0.08	0.08	0.07	0.08	0.10	0.11	0.10	0.11	0.12	0.5	14	100
	CEP63	0.07	0.06	0.11	0.11	0.13	0.12	0.10	0.09	0.09	0.5	4	33
	CEP76	0.07	0.07	0.07	0.08	0.09	0.14	0.10	0.07	0.12	0.5	12	57
	RTTN	0.09	0.09	0.09	0.10	0.10	0.11	0.09	0.09	0.08	0.0	18	14
	CEP97	0.10	0.10	0.10	0.10	0.09	0.10	0.10	0.09	0.09	-0.1	15	1
	SAS-6	0.11	0.09	0.10	0.10	0.09	0.09	0.09	0.09	0.08	-0.3	36	42
	Plk4	not detected										not detected	not detected
PCM	KIF22	0.02	0.02	0.04	0.05	0.07	0.09	0.12	0.14	0.19	3.1	33	464
	SSX2IP	0.03	0.03	0.05	0.07	0.09	0.11	0.11	0.13	0.16	2.1	20	10
	TUBG1	0.04	0.04	0.04	0.04	0.04	0.04	0.12	0.12	0.11	1.5	166	101
	tpx2	0.06	0.06	0.07	0.08	0.10	0.11	0.11	0.12	0.11	1.0	43	516
	kif11	0.06	0.06	0.07	0.08	0.09	0.10	0.10	0.11	0.12	0.9	44	185
	dync11i1	0.06	0.0735	0.0605	0.0675	0.062	0.0605	0.104	0.1185	0.087	0.8	169	38
	dync2h1	0.067	0.057	0.073	0.074	0.092	0.098	0.104	0.105	0.128	0.7	35	3
	prc1	0.06	0.07	0.08	0.09	0.10	0.12	0.09	0.10	0.10	0.7	32	13
	CEP152	0.07	0.07	0.10	0.10	0.10	0.12	0.10	0.11	0.09	0.6	7	33
	kif4a	0.08	0.07	0.08	0.09	0.09	0.09	0.10	0.10	0.10	0.4	100	120
	TUBGCP4	0.08	0.08	0.08	0.09	0.10	0.08	0.10	0.10	0.10	0.2	14	20
	nedd1	0.08	0.08	0.09	0.08	0.09	0.08	0.10	0.10	0.09	0.2	30	12
	dync2i1	0.0875	0.076	0.0915	0.088	0.0965	0.0975	0.0925	0.0935	0.101	0.1	not detected	11
	CKAP5	0.09	0.09	0.09	0.09	0.09	0.10	0.09	0.10	0.09	0.1	67	102
	dync112	0.086	0.111	0.077	0.102	0.09	0.08	0.086	0.085	0.104	0.1	411	60
	dync2i2	0.089	0.081	0.1	0.099	0.106	0.098	0.09	0.087	0.095	0.0	not detected	35
	TUBGCP3	0.09	0.08	0.09	0.09	0.10	0.10	0.09	0.10	0.09	0.0	33	10
	KATNB1	0.09	0.09	0.10	0.10	0.10	0.10	0.09	0.09	0.09	0.0	20	108
	dynl1	0.086	0.1	0.114	0.09	0.09	0.07	0.083	0.085	0.093	0.0	138	49
	dync1i2	0.091	0.104	0.104	0.073	0.081	0.106	0.091	0.089	0.092	0.0	232	48
	PCNT	0.09	0.09	0.09	0.10	0.10	0.10	0.09	0.09	0.09	0.0	16	23
	NPM2	0.08	0.10	0.08	0.12	0.08	0.08	0.07	0.07	0.09	0.0	1181	116
	dync2li1	0.086	0.074	0.084	0.086	0.089	0.091	0.082	0.081	0.091	0.0	43	4
	CLASP1	0.09	0.08	0.09	0.10	0.10	0.10	0.09	0.09	0.08	-0.1	15	48
	TUBGCP2	0.09	0.09	0.10	0.10	0.10	0.09	0.09	0.09	0.09	-0.1	59	22
	dync11i2	0.097	0.089	0.099	0.104	0.094	0.099	0.091	0.094	0.086	-0.1	26	27
	dynl1	0.099	0.101	0.094	0.096	0.092	0.105	0.089	0.093	0.089	-0.1	51	130
	NEK9	0.09	0.10	0.10	0.10	0.10	0.10	0.09	0.08	0.08	-0.2	45	27
	TACC3	0.10	0.10	0.10	0.10	0.10	0.10	0.09	0.09	0.08	-0.2	115	117
	TUBGCP6	0.10	0.09	0.11	0.10	0.10	0.10	0.08	0.08	0.08	-0.3	8	14
	NPM3	0.11	0.09	0.12	0.11	0.10	0.10	0.09	0.09	0.08	-0.3	150	70
	KATNA1	0.10	0.10	0.10	0.09	0.10	0.09	0.09	0.08	0.08	-0.3	63	30
	NPM1	0.12	0.09	0.12	0.10	0.09	0.09	0.08	0.08	0.08	-0.6	308	102
	TUBGCP5	not detected										24	not detected
	KATNAL1	not detected										5	not detected
	KATNAL2	not detected										not detected	22

Table 1.

Centrosomal components during meiotic divisions.

Expression patterns of the components of the centrioles and the pericentriolar material. The changes in protein abundance are color-coded: unchanged=white, decreased=blue, increased=light orange. The estimation of the protein and mRNA concentrations was derived by published datasets and color-coded on a white-green scale for protein concentrations and on a white-yellow scale for mRNA concentrations.

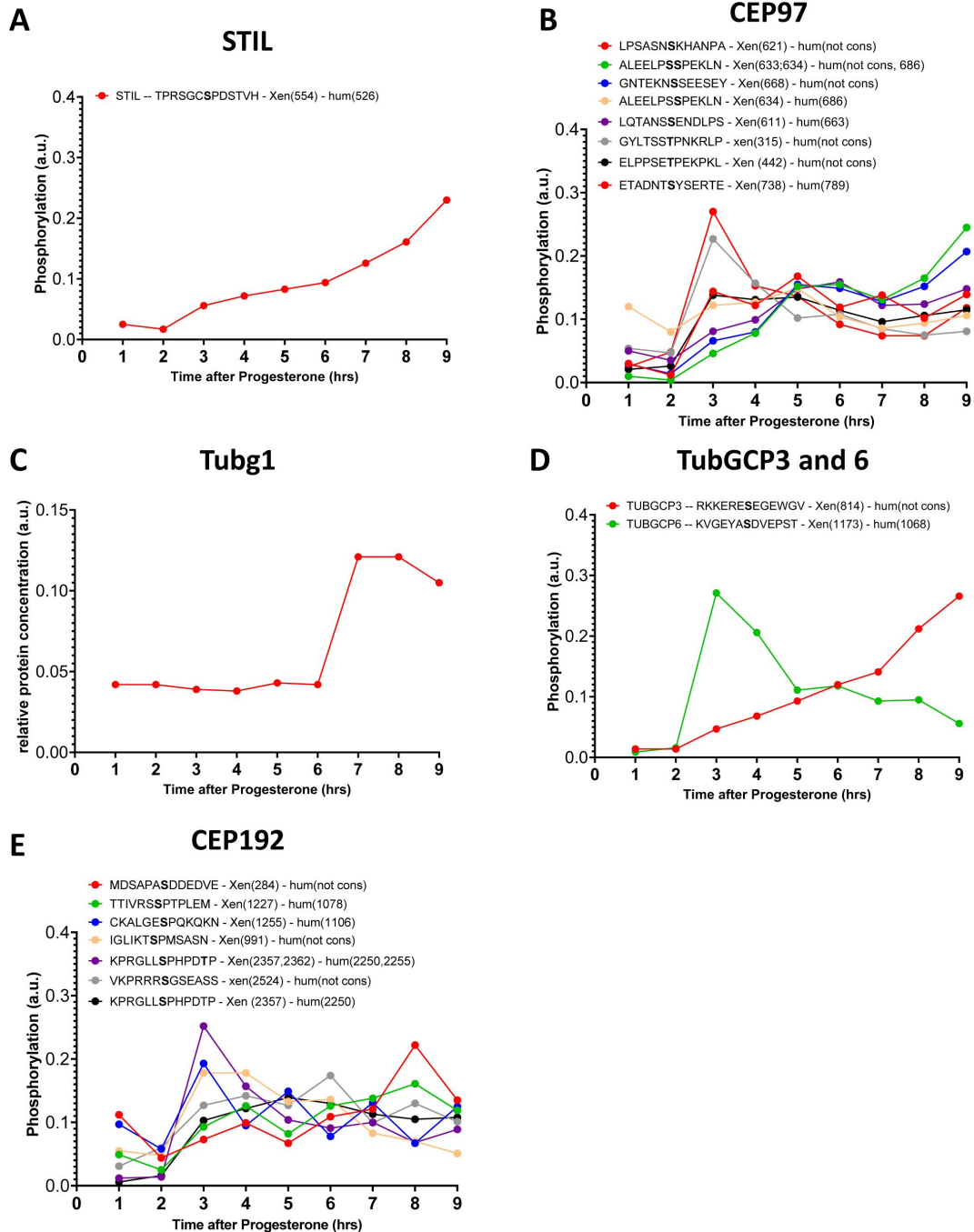


Table 1-table supplement 1.

Phosphorylation patterns of centrosomal proteins during meiotic divisions.

(A-E) The levels of phosphorylation sites of STIL (A), CEP97 (B), Tubg1 (C), TubGCP3 and 6 (D), CEP192 (E) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature.

maternal components. These include γ -tubulin (Stearns & Kirschner, 1994 [\[1\]](#); Gard, 1994 [\[2\]](#); Félix et al, 1994 [\[3\]](#)), pericentrin (Dicthenberg et al, 1998 [\[4\]](#)), Nek2B (Fry et al, 2000 [\[5\]](#); Uto, 2000 [\[6\]](#)), SSX2IP (Bärenz et al, 2013 [\[7\]](#)), TPX2, PRC1, Kif4A, Eg5/Kif11, CLASP1/Xorbits and Kif22/Xkid, CEP152 (Hatch et al, 2010 [\[8\]](#)), NEDD1 (Liu & Wiese, 2008 [\[9\]](#)), Maskin/TACC3 (Stebbins-Boaz et al, 1999 [\[10\]](#); Peset et al, 2005 [\[11\]](#)), XMAP215/CKAP5 (Popov et al, 2002 [\[12\]](#); Kinoshita et al, 2002 [\[13\]](#)) and Nercc1/NEK9 (Roig et al, 2005 [\[14\]](#)).

Most of the PCM proteins were detected in our proteomic dataset and either accumulate or stay constant during meiotic divisions (Table 1 [\[15\]](#)). Our dataset highlights that γ -tubulin accumulates during meiotic divisions at the time the oocytes enter MII (Table 1 [\[15\]](#)-table supplement 1C). GCP2 and 4 are expressed at a lower level than γ -tubulin (Table 1 [\[15\]](#)), in agreement with the expected stoichiometry of GCPs- γ -tubulin in the γ -TuRC (Sulimenko et al, 2022 [\[16\]](#)). The phosphorylation of γ -TuRC components is known to affect the stability and the activation of the complex (Sulimenko et al, 2022 [\[16\]](#)). We detected the phosphorylation of GCP3 at S814 (Class IV) and GCP6 at S1173 (Class III) (Table 1-table supplement 1D). This second phosphosite is well conserved in the human protein and targeted by Plk4, which regulates centriole duplication (Bahtz et al, 2012 [\[17\]](#)). These results indicate that the oocyte, despite the absence of centrioles, prepares a functional centrosomal machinery by accumulating and/or phosphorylating some of the PCM proteins. Apart of the γ -TuRC complex, we found that 10 other PCM proteins are phosphorylated during meiotic maturation (CEP192, SSX2IP, CEP41, TACC3/Maskin, Nucleophosmin, Katanin, Kif11, Kif15/Eg5, Numa and Dynein). CEP192 is a coiled-coil scaffolding protein that recruits γ -TuRC, Aurora-A and Plk1 to the centrosome for its maturation. Our phosphoproteome reveals that CEP192, a Class II protein, is phosphorylated at 7 sites between prophase and MI, 5 of them corresponding to Cdk1 phosphorylation sites (Table 1-table supplement 1E). Plk1 is recruited to its many different locations in the cell through its Polo-Box-Domain (PBD), which binds to phosphorylated S-S/T(p) motifs (Elia et al, 2003a [\[18\]](#), 2003b [\[19\]](#); Lee et al, 1998 [\[20\]](#); Reynolds & Ohkura, 2003 [\[21\]](#)). Two of the CEP192 sites identified in our phosphoproteome, S991 and S1227, correspond to such motifs. Therefore, by attracting Plk1, CEP192 phosphorylation could prepare the reconstitution of a functional centrosome at fertilization.

Assembly of microtubule spindles and microtubule dynamics

Xenopus prophase oocytes are inefficient in their ability to support microtubule polymerization (Jesus et al, 1984 [\[22\]](#); Gard & Kirschner, 1987 [\[23\]](#); Heidemann & Kirschner, 1975 [\[24\]](#)), although microtubules are formed at the cortical and nuclear envelope level, where γ -tubulin foci are also localized (Gard, 1994 [\[2\]](#); Jesus et al, 1988 [\[25\]](#)). The ability to assemble microtubule asters is acquired at the time of NEBD, when a giant disk-shaped MTOC generates a large microtubular monoaster at the basal region of the disintegrating nucleus (Huchon et al, 1981 [\[26\]](#); Gard, 1994 [\[2\]](#)). Therefore, MTOC and microtubule-associated proteins (MAPs) must be finely regulated by post-translational modifications at time of NEBD. The transient microtubule monoaster transports the meiotic chromosomes to the animal cortex, where the first meiotic spindle is assembled (Huchon et al, 1981 [\[26\]](#); Gard, 1992 [\[27\]](#)). Due to the absence of centrioles, meiotic spindles are barrel-shaped and are formed by centrosome-independent microtubule nucleation originating from the chromosomes and involving chromatin and the Ran GTPase (Gruss & Vernos, 2004 [\[28\]](#)), much like spindles of vascular plant cells. Moreover, the two successive spindles are differentially regulated. The MI spindle is a transient and dynamic structure while the MII spindle is stable for hours, anchored to the plasma membrane until fertilization (Gard, 1992 [\[27\]](#)). Therefore, it is possible that MAPs are specifically and differentially regulated at MI and MII (Classes III and V). Finally, a major change in microtubule assembly occurs during the transition from meiosis to embryonic mitosis. The assembly of the first embryonic spindle integrates the meiotic chromosome-dependent pathway with the newly formed centrosomes (Cavazza et al, 2016 [\[29\]](#)). Thereafter, subsequent embryonic spindles depend solely on the microtubule nucleating activity of the centrosomes (Kirschner, 1986 [\[30\]](#)). All these critical transitions in the way of assembling a microtubular spindle depend on the proteins stored in the oocyte.

Many regulators of the microtubular spindle are highly regulated during meiotic divisions, both at level of their accumulation (**Figure 2B-C**) and by post-translation modifications (**Figure 8**). At least 5 microtubule regulatory proteins undergo specific patterns of phosphorylation that deserve attention (**Figure 8**). Katanin (KATNB1), which is responsible for the majority of M-phase severing activity in *Xenopus* eggs and is activated by Cdk1 (McNally & Thomas, 1998), displays a bi-modal pattern of phosphorylation, increasing at the time of formation of the monoaster-MTOC and MI spindle, decreasing during MI-MII transition and increasing again at MII (**Figure 8**). Kif11/Eg5 phosphorylation occurs at T936 and T937 starting NEBD (**Figure 8**). Both phosphosites were reported in a broad range of models, and the second site was shown to be phosphorylated by Cdk1 (Blangy et al, 1995; Cahu et al, 2008; Giet et al, 1999) (**Figure 8**). Kif11/Eg5 controls the relative stability of bipolar versus monopolar organization of spindles in *Xenopus* egg extracts (Mitchison et al, 2005) but also centrosome disjunction and/or separation. Hence, Kif11 is a good candidate to be an important player regulating the original changes in microtubule dynamics and organization of the two successive spindles. Kif15, also known as XKlp2, a plus end-directed kinesin protein required for centrosome separation and maintenance of spindle bipolarity in *Xenopus* egg extracts (Boleti et al, 1996), is phosphorylated at S681 and S692 during meiotic maturation. S681 is rather phosphorylated at MI while S692 is specific of MII (**Figure 8**), suggesting a potential distinct regulation of Kif15 on microtubule dynamics at MI and MII. In contrast to Kif11/Eg5 and Kif15, some kinesins from the kinesin-14 family, such as Kifc1 whose protein level increases during meiotic divisions (**Figure 2C**), as well as dyneins, are motor proteins that move to microtubules in the opposite direction to kinesins. Among the dynein family, the two paralog genes dynein light intermediate chain (DynC1LI1 and DynC1LI2) play important roles in mitosis, including positioning the spindle and focusing the MTs into poles. Our phosphoproteome reveals multiple phosphorylations of these proteins, starting in MI and remaining constant or progressively increasing until MII, as for S197 of DynC1LI1 (S207 in human) (**Figure 8**). This agrees with previous data showing that S197 is phosphorylated by Cdk1 (Dell et al, 2000; Addinall et al, 2001) and that DynC1LI becomes hyperphosphorylated at the time of NEBD and remains hyperphosphorylated throughout the rest of meiosis (Huang et al, 1999). Additionally, our phosphoproteome reveals that 3 sites (495, 498 and 501) are phosphorylated around NEBD and get dephosphorylated after MI (Class III), hence being correlated with the 1st meiotic division and not the 2nd one, while 6 other sites (197, 396, 409, 446, 452, and 467) are phosphorylated both in MI and MII (Class II) (**Figure 8**). Dynein is required for organization of the original microtubule/MTOC array that is organized at NEBD, together with XMAP215, XKCM1 and Numa (Becker et al, 2003). The formation of this oocyte original monoaster/MTOC could involve the dynein phosphorylations that transiently take place precisely at this period.

Strikingly, MI and MII spindles segregate chromosomes and chromatids in a very different way. In MI, the homologous chromosomes are segregated, whereas in MII the sister chromatids are displaced in the daughter cells, due to a specific organization of kinetochores and regulation of cohesins. In MI, arm cohesins are phosphorylated and degraded whereas centromeric cohesins are protected from cleavage by Sugoshin (Sgo1)-PP2A. This allows the separation of chromosomes, but chromatids are still attached together. In MII, centromeric cohesins are fully degraded, allowing the segregation of sister chromatids, as during mitosis (Wassmann, 2022). How a fraction of cohesins is protected from cleavage in MI but not in MII is still not entirely clear since the phosphorylation of cohesins and their partners, which are not very abundant proteins, is difficult to be detected with conventional shotgun phosphoproteomics. Notably, our phosphoproteomic studies highlight new sites differentially phosphorylated between MI and MII within these proteins. Sgo1 and two components of the cohesin complex, Pds5b and Wapl, are increasingly phosphorylated during meiotic divisions (Class IV) (**Table 1**-table supplement 1E). Noteworthy, S1069 of Wapl is a putative Plk1 site (Grosstessner-Hain et al, 2011; Kettenbach et al, 2011). By highlighting new sites differentially phosphorylated between MI and MII, our phosphoproteomic studies provide clues to elucidate the still unknown mechanisms of the meiotic mechanism of cohesion protection-deprotection.

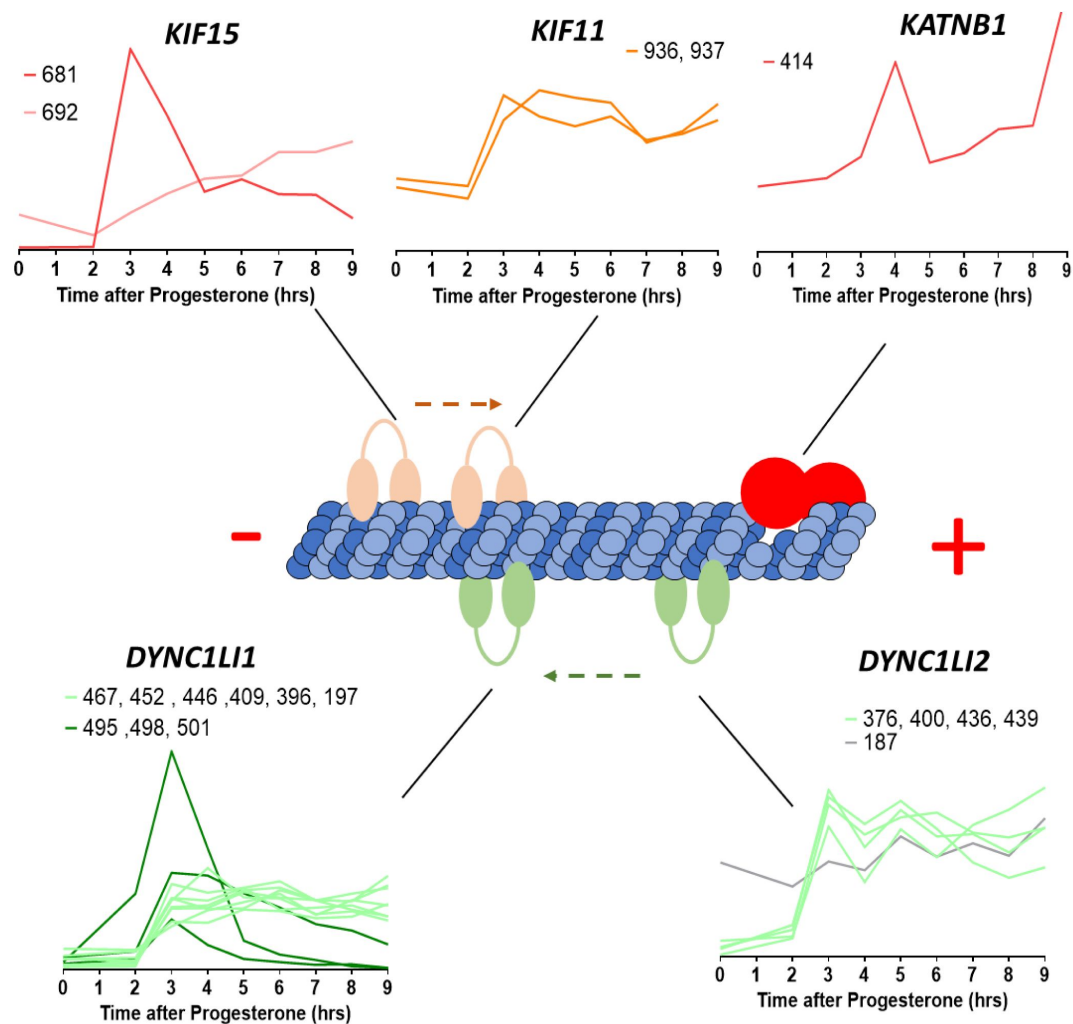


Figure 8.

Microtubule-interacting proteins phosphorylation during meiotic divisions.

Graphic representation of the phosphorylation dynamics of kinesins (Kif15, Kif11), dyneins (DYNC1LI1, DYNC1LI2) and katanin (KATNB1) during meiotic divisions. The sites of the phosphorylation detected are marked above each panel.

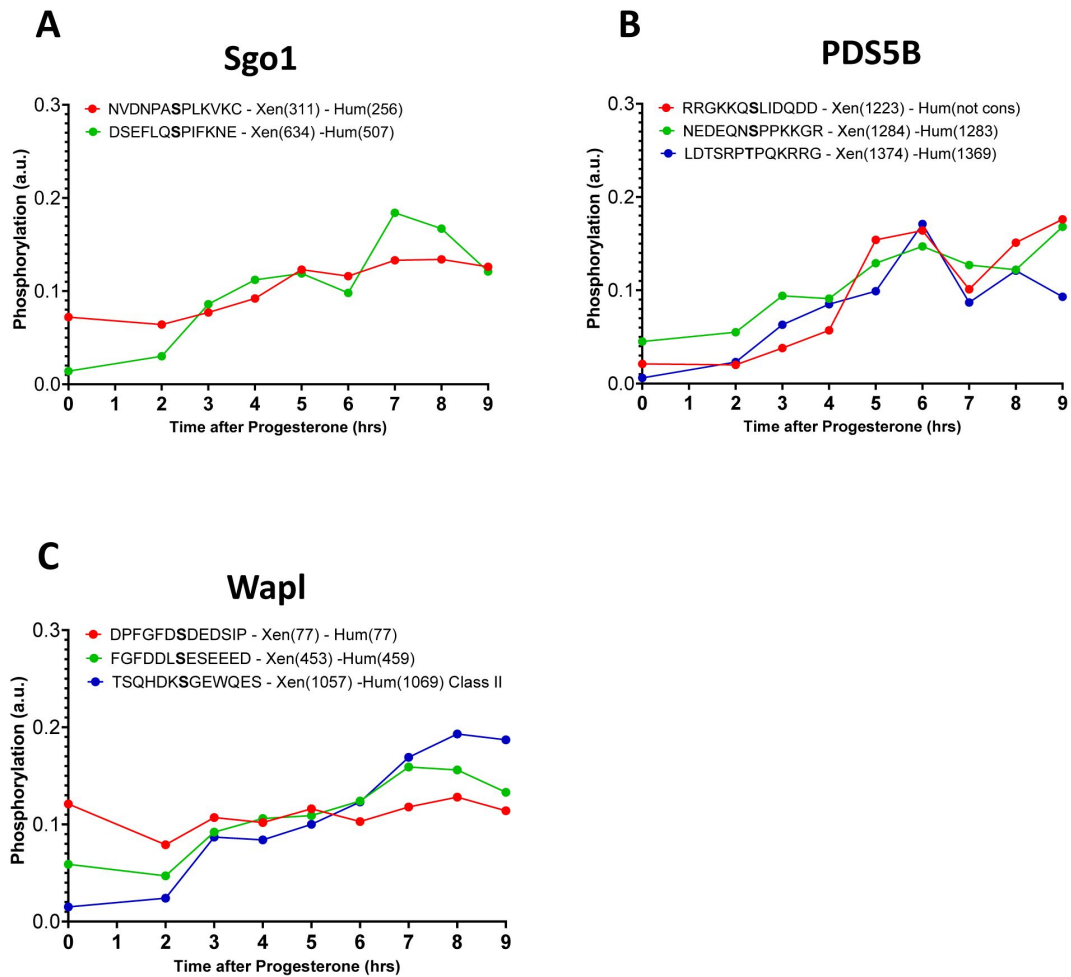


Figure 8-figure supplement 1.

Phosphorylation patterns of cohesins during meiotic divisions.

(A-C) The levels of phosphorylation sites of Sgo1 (A), PDS5B (B) and Wapl (C) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The position of the residue phosphorylated is reported in brackets according to the *Xenopus* and human nomenclature.

Evolutional conservation of the phosphoproteome between *Xenopus* and mouse

We compared our phosphoproteome to a dataset obtained during meiotic divisions in mouse oocytes (Sun et al, 2024¹). We identified 408 phosphorylation sites present in both datasets (see Methods) (Appendix 7). These sites correspond to 320 *Xenopus* proteins and 277 mouse proteins. Note that when multiple mouse proteins share the same phospho-peptide sequence, they are grouped together in the mouse dataset. However, alignment with *Xenopus* data allows us to resolve which specific mouse protein is phosphorylated, thereby enriching the mouse dataset by leveraging information from *Xenopus* measurements. We then assessed whether the changes in phosphorylation between prophase and metaphase II were consistent between *Xenopus* and mouse meiotic maturation (**Figure 9A**¹). Interestingly, the phosphorylation dynamics were significantly correlated between the two datasets (Pearson coefficient: 0.39, $p < 0.0001$) (**Figure 9A**¹). Furthermore, the changes in phosphorylation observed in *Xenopus* oocytes (range of the Log₂ Fold Change (MII/Pro): -4.27 and 7.03) were substantially greater than those measured in the mouse dataset (range of the Log₂Fold Change(MII/Pro): -2.55 and 3.72), likely reflecting the higher dynamic range achieved by our workflow. Among the key phosphorylation events detected in both datasets are those involved in the activation of Cdk1 and the Mos/MAPK pathway. Notably, phosphorylation of Plk1 at T210 (mouse numbering, T201 in *Xenopus*), which is catalyzed by Aurora A and is critical for Plk1 activation (Macûrek et al, 2008²), as well as phosphorylation of Gwl at S442 (mouse numbering, 467 in *Xenopus*), increased in both datasets (**Figures 6B**¹ and **9A-B**¹). Similarly, both datasets consistently detected phosphorylation of Erk2 at T183 (mouse numbering, T188 in *Xenopus*), which reflects MEK-dependent activation of this kinase (Payne et al, 1991³)(**Figures 6B**¹ and **9A-B**¹). Among the conserved sites whose phosphorylation increases during meiosis, several are known targets of Cdk1 and Erk/MAPK (**Figure 9B**¹), consistent with the activation of these kinases during meiotic maturation. We experimentally confirmed the phosphorylation of Fak1 at S910, Erk2 at T188, and Plk1 at T201 by western blot (**Figure 9C**¹). Moreover, by microinjecting the Cdk1-specific inhibitor Cip1 in oocytes in order to prevent Cdk1 activation, we demonstrated that Cdk1 is required for phosphorylation of all these substrates, whether they are direct targets of Cdk1 or Erk/MAPK (**Figure 9C**¹). This aligns with previous reports indicating that MAPK pathway activation in oocytes occurs downstream of the initial activation of Cdk1 (Santoni et al, 2024⁴). Interestingly, AKTS1/PRAS40 phosphorylation decreases during meiotic maturation in both species (**Figure 9A-B**¹). We experimentally validated the dephosphorylation of AKTS1/PRAS40 at S208 and showed that this event occurs at NEBD and depends on Cdk1 activity (**Figure 9C**¹). AKTS1/PRAS40 is a direct inhibitor of mTORC1 by competing with 4E-BP1 and S6K1 for binding to raptor (Wang et al, 2007⁵; Oshiro et al, 2007⁶). AKTS1/PRAS40 is phosphorylated on S184 (mouse numbering, S208 in *Xenopus*) by mTORC1, which leads to its dissociation from mTOR and enhances its ability to phosphorylate its substrates (Oshiro et al, 2007⁶). Interestingly, previous studies suggest that translation of ribosomal proteins, which are targets of mTOR signaling via the TOP motif located in their 5'-UTR, decreases during meiotic maturation (Luong et al, 2020⁷). Our findings support these observations and provide additional evidence for the downregulation of mTOR activity upon meiotic resumption.

Discussion

The amphibian oocyte and egg have long been a source of inspiration and a source of experimental opportunity for important areas of biology, such as nuclear organization, spindle formation, cell polarity, fertilization, mitosis and meiosis, endocrine signaling and intracellular signal transduction. As experimental embryology gave way to genetic approaches in the late 20th century to study embryonic development, the attractiveness of amphibian oocytes declined.

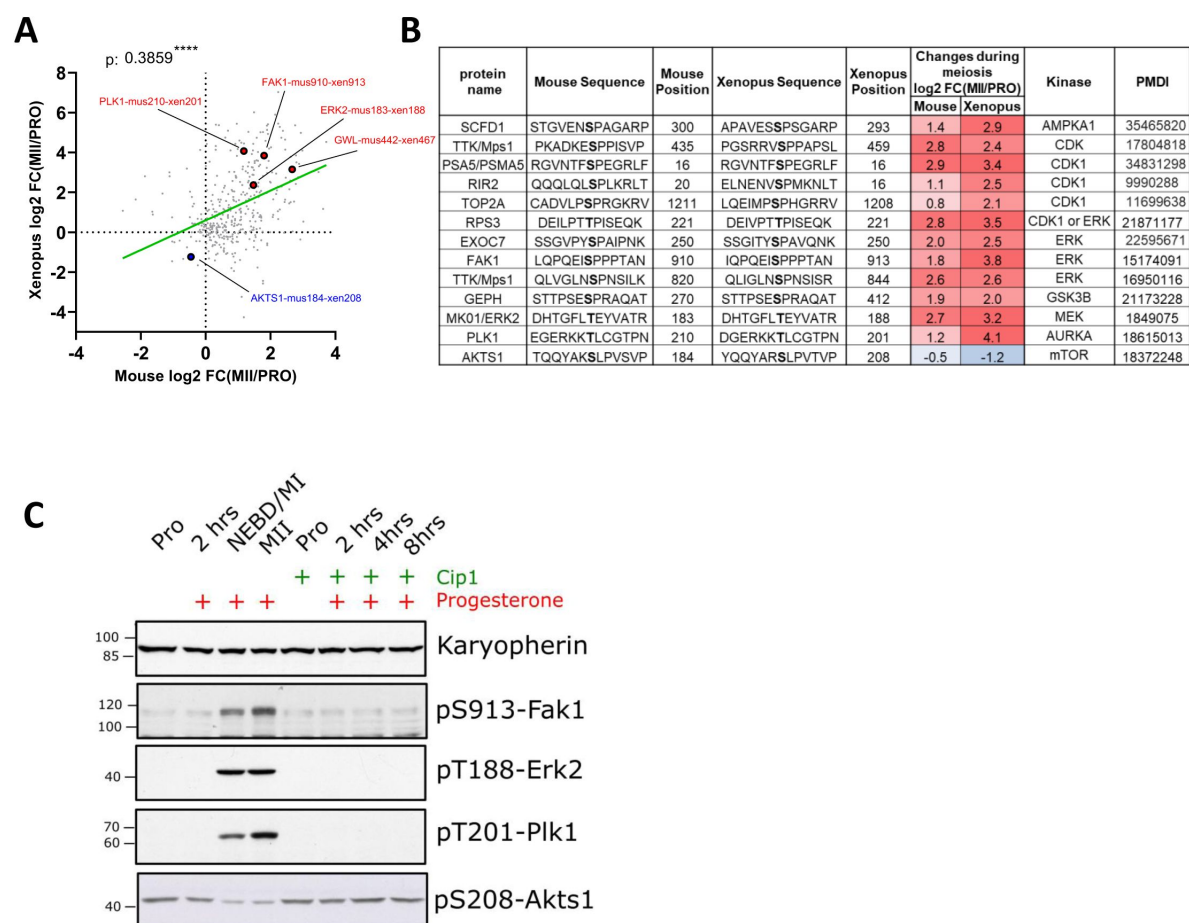


Figure 9.

(A) Comparison of phosphorylation changes during meiotic divisions in *Xenopus* (our dataset) and mouse oocytes (Sun et al., 2024). Log2 fold changes in phosphorylation between prophase (PRO) and metaphase II are plotted. The Pearson correlation ($r = 0.39$, $p < 0.0001$) was calculated. **(B)** Selected examples of conserved phosphorylation sites between mouse and *Xenopus*. The kinase that was experimentally identified to phosphorylate the specific site is displayed together with the supporting reference. **(C)** Oocytes were microinjected with a specific inhibitor of Cdk1, Cip1. After overnight incubation, meiotic maturation was induced by progesterone. Oocytes were collected at different times after progesterone treatment. Karyopherin was used as a loading control. The phosphorylation of Fak1 at S913, Erk2 at T188, Plk1 at T201, and Akts1 at S208 was evaluated by western blot.

The original advantages of the amphibian oocyte are now becoming more obvious, as it becomes important to link complex cellular processes, like cell division, to signal transduction. These processes are difficult to study because they occur at a single cell or even at subcellular level. Indeed, many important cell-division related processes involve transient changes occurring in specific compartments of the cell, like the endoplasmic reticulum, the mitotic apparatus, the nuclear and plasma membranes, or the cytoskeleton. Additionally, meiotic and mitotic divisions occupy a very short fraction of the life span of germ and somatic cells, making it hard to study them in cultured cells. Frog oocytes and eggs show exquisite synchrony of cell division, allowing to have large cohort of cells at the same stage of the cell cycle. Therefore, we and others have returned to the awesome power of the amphibian oocyte and egg to study the signal transduction, intracellular posttranslational changes surrounding the onset of meiotic and mitotic divisions and fertilization at a single cell level.

Importantly, the ingenious and sensitive tools developed to study the transcriptional regulation are of little use in the study of a cell type that does not transcribe RNA and of events that are regulated post-translationally, as meiotic and early embryonic divisions. For a long time, we lacked general tools for identification and measurement of the activity of the regulators of these processes, impairing our ability to connect the morphological changes previously described by histology to the respective molecular events. One manner to investigate these pathways is using pharmacological inhibitors, but their non-specificity and the complexity of pathways pose serious limitations. For years, phosphorylation events have been extensively studied using candidate-based approaches, as site-specific mutagenesis and phosphospecific antibodies. These targeted-approaches have several caveats. The selection of the putative sites of phosphorylation is based on *in silico* evidence, limiting the analysis to proteins bearing canonical consensus sequences and hence our knowledge of novel phosphorylation motifs. When multiple putative phosphorylation sites are present in the same protein, mutagenesis of individual sites or combinations of sites becomes too complex, impairing identification of the phosphorylation sites used *in vivo* and their interdependency or redundancy. Moreover, the time-consuming production of phospho-specific antibodies is not scalable to high throughput applications. Mass spectrometry has the potential of shedding light on these intricate networks. However, the current tools are unable to accurately quantify the level of proteins and phosphorylated proteins at the single cell level -except for the frog oocytes and eggs, thanks to their exceptional size and high protein content.

Oocyte meiotic maturation is a key biological process. First, it transforms the oocyte into a fertilizable cell at the origin of life of any animal. Second, since meiosis derives from mitosis, thanks to cellular adaptations that were necessary to produce a haploid gamete (preventing an S-phase between the two divisions, modifying the functioning of the spindle during meiosis I, establishing division arrests needed for the oocyte to grow or to wait for sperm, performing asymmetrical divisions to save nutrient reserves, etc.), its study provides the opportunity to understand how cell division can adapt and diversify in living systems, depending on specific cellular organization and specialized functions.

In this paper, we have applied the current state of the art of quantitative mass spectrometry to oocyte maturation and meiotic divisions using quantitative proteomics and quantitative phosphoproteomics. We have temporally correlated events with oocyte maturation induced naturally by progesterone to *Xenopus* full-grown oocytes and watched the progression through meiosis I to the natural arrest at the second meiotic metaphase. We have been able to correlate the proteomic measurements with the physiological changes as the oocyte, as it goes from an early prophase I state through to a second meiotic metaphase state. We find dramatic changes in the oocyte nucleus and the meiotic spindles and drastic modifications to the standard cell cycle. These dramatic changes are choreographed principally by signaling pathways employing kinases and phosphatases. In all animals, meiotic maturation relies on the dynamics of phosphorylation events, either during the signaling pathway that leads to Cdk1 activation or during the Cdk1 downstream period that orchestrates the structural events of the two meiotic divisions. The

candidate-based approaches have provided a global scheme of this process based on the knowledge of a few kinases that have been extensively studied for years. However, this scheme still does not provide a very comprehensive picture of the meiotic process, since it does not take into account hundreds of unknown critical kinases, substrates and phosphosites. Our quantitative proteomics and phosphoproteomics approach allowed to obtain a much deeper and comprehensive dynamic picture of the phosphorylated sites during meiotic maturation. This has enabled us to describe and support with biochemical precision the various processes in the oocyte, determine the relative timing of various site phosphorylations and molecular processes, thereby challenging or suggesting the epistatic sequences of specific molecular pathways. Furthermore, we confirmed the importance of certain previously known phosphosites, identified the likely kinases or in some cases the kinase families responsible for substrate phosphorylation, and established connections between phosphorylation events and intracellular remodeling, as well as regulatory mechanisms such as translation and protein degradation.

In addition to explaining the order of diverse events of chromatin remodeling, nuclear organization, plasma membrane transformations, cytoskeletal changes of meiosis, our proteomic and phospho-proteomic datasets should provide a large number of experimental avenues for exploring similar events in both mitotic and meiotic stages of diverse organisms. This molecular data will enable us to better understand regulatory pathways involving chromosome behavior, spindle formation, nuclear and plasma membrane regulation, and signal transduction and derive insights far beyond the scope of meiosis and oocyte maturation. One such insight might come from a closer analysis of the quality control mechanisms in the oocyte. Both nucleopore remodeling (King et al, 2019 [↗](#)) and E2/E3 ubiquitin ligases are involved in housekeeping and quality control, specifically in the removal of aggregated and orphan complex proteins, as the germ cell of an organism prepares to become the next generation

Methods

Oocytes collection and *in vitro* maturation

The research with *Xenopus laevis* was performed under the oversight of the Harvard Medical Area Institutional Animal Care and Use Committee. Mature *Xenopus laevis* females were used after priming with pregnant mare serum gonadotropin (PMSG, 50 IU per frog). A sample of ovary tissue was surgically removed, rinsed in 1× Marc's modified Ringer's (MMR) and placed in OR2 buffer (83 mM NaCl, 2.5 mM KCl, 1 mM CaCl₂, 1 mM MgCl₂, 1 mM Na₂HPO₄, 5 mM HEPES, pH 7.8) and oocytes were manually defolliculated. Prophase oocytes were selected and kept in dishes with agar beds to avoid sticking to the dish. For the Cip1 experiment, prophase oocytes were microinjected with 60 ng of GST-XenCip1, as previously described (Santoni et al, 2024 [↗](#)). Maturation was induced by incubation at 18 °C with 10 µg/mL progesterone in OR2 buffer.

Oocytes were flash frozen with liquid nitrogen in bunches of ten or fifty for protein and phospho - proteomics measurements respectively. All samples were taken in triplicates, each biological replicate was performed with different frogs. The first sample was taken from untreated oocytes at the time of progesterone treatment, the second timepoint 2 hours after progesterone treatment, with hourly timepoints past that, up to 9 hours post treatment.

Oocyte lysis and digestion into peptides

For each condition the oocytes were lysed at a ratio of 6 µL buffer/embryo in the same buffer as (Presler et al, 2017 [↗](#)) (0.25 M sucrose, 1% replacement NP-40, 10 mM EDTA, 25 mM HEPES, 1 mM TCEP, 10 µM Combretastatin, 10 µM Cytochalasin D, Roche Protease Inhibitor Mini EDTA free, and Phosstop inhibitor (Roche) at pH 7.2). The yolk was removed by spinning the samples at 4,000 g for 4 minutes at 4°C. Both the supernatant and all lipids were transferred for a new tube. The concentration of HEPES was raised to 0.1 M and SDS was added to 20%. After addition of DTT to 5

mM, the samples were denatured in a 60°C water bath for 20 minutes. After the samples temperatures return to room temperature, N-Ethylmaleimide (NEM) was added to 15 mM, tubes were mixed thoroughly, and the reaction proceeded for 1 hour. Then additional DTT was added to 5 mM to quench the excess NEM. The samples were flash frozen and stored at -80°C until the methanol-chloroform precipitation (Wessel & Flügge, 1984 [↗](#)). Protein was precipitated by methanol-chloroform precipitation. Due to the large volume of lysate (~ 1 mL), the precipitation was performed in glass 15 mL Corex tubes (No. 8441). The tubes were cleaned for a two-hour chromic acid incubation and rinsed extensively in filtered water. Then they were autoclaved and before use for the precipitation, they were rinsed with HPLC H₂O. The centrifugations were done in a JA 50 30 Ti rotor in a Beckman Avanti J-30I centrifuge at 21,000 g for either 6 minutes or 15 minutes. After the last supernatant was removed, the pellets were air dried for 15 minutes and resuspended in 0.6 mL of 6 M guanidine, 50 mM EPPS pH 8.5 and transferred to a 2 mL Eppendorf tube. The samples were incubated for 5 minutes at 65°C and the protein amount was determined by a BCA Assay. The solution was diluted to 2 M guanidine with 10 mM EPPS pH 8.5. Lysyl endopeptidase (FUJIFILM WAKO (129-02541) resuspended with HPLC H₂O to ~2 mg/mL) was added to a concentration of 20 ng/mL and the initial digestion proceeded for 12 hours at room temperature with gentle shaking. The samples were further diluted to 0.5 M guanidine with 10 mM EPPS pH 8.5 and Trypsin protease (Sequencing Grade Modified Trypsin, Promega) was added to a final concentration of 10 ng/mL along with an additional 20 ng/mL of LysC. The samples were incubated in a 37°C room for 8 hours on a nutator.

Phospho-peptide enrichment

To carry out the phosphoproteomic analysis, we made the conventional steps of digesting the proteins from each time sample with trypsin and specifically labeling them with isobaric tags (TMT-10 reagents). Tagged samples and the phosphor-peptides were enriched on an IMAC column. Importantly, we chose to multiplex peptides before the phosphor-peptide enrichment to improve data quality. There is a tradeoff of decreased yield and therefore depth, as it is not economical to label more than a few milligrams of material. We used 2.5-4 mgs of TMT-labeled peptides per replicate, enriched with 5 µm Titanium Dioxide microspheres (GL Sciences 5020-75000) and fractionated as previously described (Presler et al, 2017 [↗](#)). A typical yield of 50-80 µg of peptides eluted from the column, with a median phosphor-peptide enrichment of ~80%.

Phosphatase treatment of *Xenopus* samples prepared for parallel phospho-enrichment

We adapted the method that we used to phosphatase-treat digested peptides (Presler et al, 2017 [↗](#)) for the larger samples used here for the parallel phospho-enrichment. We chose to phosphatase treat when proteases were present in the samples (but after the protease digestion sequence described above) to enable the de-phosphorylated peptides to have the cleavage pattern expected if they were present initially in the de-phosphorylated state. We used the heat-labile Alkaline Shrimp Phosphatase (Affymetrix Inc, USA). We used a 3k Amicon Centrifugal Filter to exchange the enzyme storage buffer to 10 mM EPPS pH 8.5 instead of Tris-HCl pH 7.5. We also concentrated the phosphatase to between 3-4 U/µL (confirmed by p-nitrophenyl phosphate assay). We added an estimated 833 units of phosphatase to two of the samples built to contain representative mixture of peptides (TMT 131N, channel 10: 100 oocytes collected at the latest timepoint, and TMT 131C, channel 11: 50 oocytes collected at the first timepoint combined with 50 oocytes collected at the last timepoint). To samples that were not phosphatase-treated, we added the same volume of blank phosphatase buffer: 5 mM EPPS pH 8, 50% glycerol. To all samples, we added MgCl₂ to 10 mM and EDTA to 0.1 mM (to chelate metals that might compete with the magnesium necessary for enzymatic activity). All the samples were incubated with gentle shaking at room temperature for 12 hours. The phosphatase was then inactivated by incubation with gentle shaking in an air-incubator at 65°C for 15 minutes. Peptide fractionation and TMT-MS3 LC/MS were carried out as done previously (Presler et al, 2017 [↗](#)).

Computing phospho-occupancy systematically

To compute phospho-occupancy, we used the `phospho_occupancy_matlab` package, available as part of our code repository (<https://github.com/elizabeth-van-itallie>) implementing the methods previously described (Van Itallie et al, 2025). We grouped phospho-peptides by unique phosphosite identifiers. In order to rigorously reflect the measurement accuracy and to take advantage of many measurements of the same phosphosite (including on peptides with different missed-cleavages or oxidized methionines), we used BACIQ (Peshkin et al, 2019) to compute the estimated trend with confidence intervals. Then, we created phosphosite sets to connect phosphosites on residues that could be on the same peptide. We made these assignments based on whether the individual phosphosites were ever measured with another phosphosite as a composite phosphosite. In order to determine occupancy, we needed to measure the phosphosite as a different form (usually not-phosphorylated). To do this, we identified non-phosphorylated peptides measured from the same experimental conditions that include the residue(s) of the phosphosite. Again, multiple non-phosphorylated peptides can include the residues of interest and the same peptide sequences can be measured multiple times. Therefore, we again use BACIQ (Peshkin et al, 2019) to aggregate these measurements in a trend with confidence intervals.

Comparison of mouse and *Xenopus* phosphosites

To compare the dynamics of mouse and *Xenopus* phosphosites between two species, we used the `phospho_matching` package, available as part of our code repository (<https://github.com/elizabeth-van-itallie>) implementing the methods previously described (Van Itallie et al, 2025). We first matched the phosphosites using the protein sequence and phosphorylated residues, then filtered the list to only keep matched phosphosites measured in both datasets. For cross-species phosphomatching, we used the default parameters that are defined in the methods section in (Van Itallie et al, 2025): BLOSUM90 substitution penalty matrix and blastp alignment results with E-value less than 1e-20.

Mass spectrometry data mapping and analysis

Peptide-Spectra matches were performed as previously described (Sonnett et al, 2018). Assignment of MS2 spectra was performed using the SEQUEST (Eng et al, 1994) algorithm by searching the data against the appropriate proteome reference set acquired from Xenbase (*Xenopus laevis* assembly 9.1) along with common contaminants: human keratins and trypsin. The target-decoy strategy was used to construct a second database of reversed sequences that were used to estimate the false discovery rate on the peptide level. SEQUEST searches were performed using a 20 ppm precursor ion tolerance with the requirement that both N- and C-terminal peptide ends are consistent with the protease specificities of LysC and Trypsin. TMT (+229.162932 Da) was set as a static modification on N-termini and lysines peptides, and N-ethyl maleimide (+125.047679 Da) was set as a static modification on cysteine residues. Oxidation of methionine (+15.99492 Da) was set as a variable modification. A peptide level MS2 spectral assignment false discovery rate of 1% was obtained by applying the target-decoy strategy with linear discriminant analysis. Peptides of 7 amino acids length or longer were ranked by linear discriminant analysis score and were filtered to a ratio of 1% reverses/forwards + reverses. Peptides were assigned to proteins and a second filtering step to obtain a 1% FDR on the protein level was applied. Peptides that matched multiple proteins were assigned to the proteins with the most unique peptides. We only used Isolation Specificity > 0.75 spectra.

Estimation of the absolute abundance of proteins

The absolute protein concentration was estimated according previously published methods (Wühr et al, 2014; Peshkin et al, 2019), based here on ion current prorated to the isobarically labeled fractions. The respective sample fractions were estimated using our previously published Bayesian

approach (Peshkin et al, 2019 [↗](#)), which integrates peptide signal and peptide-level measurement agreement into a maximum likelihood estimate of the true protein ratio and the associated confidence interval (Peshkin et al, 2019 [↗](#)). See Appendix Supplementary Methods.

Western blot and Antibodies

Oocyte homogenization and SDS-page was performed as previously described in (Santoni et al, 2024 [↗](#)). The following antibodies were used: Karyopherin (1:2000, goat, Santa Cruz sc-1863), pT188-pY190-Erk2 (1:2000, mouse, Cell Signaling 9106), pT201-Plk1 (1:1000, Abcam Ab39068), pS910-FAK (1:5000, rabbit, Invitrogen 4459G) and pS183-Akts1/PRAS40 (1:5000, rabbit, Cell Signaling 5936). The appropriate secondary HRP-coupled antibodies were used at 1:10000 dilution (Jackson ImmunoResearch).

Data availability

The mass spectrometry proteomics raw data and reference set of sequences have been deposited to the ProteomeXchange consortium via the PRIDE partner repository with the dataset identifier MassIVE MSV000094498. In addition, we developed a multi-functional Web portal to release the data and enable interactive interrogation (https://xenopus.hms.harvard.edu/index_oocyte.html [↗](#)). All of the aspects from overall protein trends to individual clusters to specific phosphosites and stoichiometry information are easily available. The portal provides per-protein link to external information about each human protein by symbol from GeneCards (<https://www.genecards.org/> [↗](#)) and respective *Xenopus* protein at Xenbase (<https://www.xenbase.org/xenbase/> [↗](#)).

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Additional information

Author contributions

Leonid Peshkin: Conceptualization; Methodology; Experiments; Software; Formal Analysis; Data Curation; Writing; Supervision. **Enrico M Daldello:** Formal Analysis; Experiments; Writing; Supervision. **Elizabeth Van Itallie:** Software; Formal Analysis; Data Curation; Writing. **Matthew Sonnett:** Methodology; Formal Analysis. **Johannes Kreuzer:** Formal Analysis; Data Curation. **Wilhelm Haas:** Methodology; Formal Analysis. **Marc W Kirschner:** Conceptualization; Methodology; Experiments; Writing; Supervision. **Catherine Jesus:** Writing; Formal Analysis; Supervision. **Marc W Kirschner** and **Catherine Jesus** contributed equally to this work.

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Author information

Leonid Peshkin[#]

Systems Biology Department, Harvard Medical School, Boston, United States
ORCID iD: [0000-0002-6420-848X](#)

#Equal contribution

Enrico Maria Daldello[#]

Sorbonne Université, CNRS, Laboratoire de Biologie du Développement - Institut de Biologie
Paris Seine, LBD - IBPS, Paris, France

ORCID iD: [0000-0002-0456-8950](https://orcid.org/0000-0002-0456-8950)

#Equal contribution

Elizabeth Van Itallie

Systems Biology Department, Harvard Medical School, Boston, United States

ORCID iD: [0000-0002-0736-7345](https://orcid.org/0000-0002-0736-7345)

Matthew Sonnett

Systems Biology Department, Harvard Medical School, Boston, United States

ORCID iD: [0000-0002-6795-1308](https://orcid.org/0000-0002-6795-1308)

Johannes Kreuzer

Systems Biology Department, Harvard Medical School, Boston, United States

ORCID iD: [0000-0003-4716-2843](https://orcid.org/0000-0003-4716-2843)

Wilhelm Haas

Systems Biology Department, Harvard Medical School, Boston, United States

Marc W Kirschner

Systems Biology Department, Harvard Medical School, Boston, United States

Catherine Jessus

Sorbonne Université, CNRS, Laboratoire de Biologie du Développement - Institut de Biologie
Paris Seine, LBD - IBPS, Paris, France

ORCID iD: [0000-0001-9296-6336](https://orcid.org/0000-0001-9296-6336)

Editors

Reviewing Editor

Akira Shinohara

Osaka University, Suita/Osaka, Japan

Senior Editor

Adèle Marston

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Reviewer #1 (Public review):

In the revised version of the manuscript, the authors have adequately addressed all our concerns. The authors should spell check their manuscript, e.g., correct phosphor-site to phospho-site etc.

Summary:

The study aims to create a comprehensive repository about the changes in protein abundance and their modification during oocyte maturation in *Xenopus laevis*.

Reviewer #2 (Public review):

Summary:

The authors analyzed *Xenopus* oocytes at different stages of meiosis using quantitative phosphoproteomics. Their advanced methods and analyses revealed changes in protein abundances and phosphorylation states to an unprecedented depth and quantitative detail. In the manuscript they provide an excellent interpretation of these findings putting them in the context of past literature in *Xenopus* as well as in other model systems. The clarity of these explanations improved significantly in the revised version of the manuscript, and several minor imprecisions have been corrected as well.

Strengths:

High-quality data, careful and detailed analysis, and outstanding interpretation in the context of the large body of literature.

Weaknesses:

Merely a resource, none of the findings are tested in functional experiments.

I am very impressed by the quality of the data and the careful and detailed interpretation of the findings. In this form, the manuscript will be an excellent resource to the cell division community in general, and it presents a very large number of hypotheses that can be tested in future experiments. *Xenopus* has been and still is a popular and powerful model system that led to critical discoveries around countless cellular processes, including the spindle, nuclear envelope, and translational regulation, just to name a few. This also includes a huge body of literature on the cell cycle describing its phosphoregulation. It is indeed somewhat frustrating to see that these earlier studies using phospho-mutants and phospho-antibodies were just scratching the surface. The phosphoproteomics analysis presented here reveals much more extensive and much more dynamic changes in phosphorylation states. Thereby, in my opinion, this manuscript opens a completely new chapter in this line of research, setting the stage for more systematic future studies.

<https://doi.org/10.7554/eLife.104255.2.sa2>**Reviewer #3 (Public review):**

Summary:

The authors performed time-resolved proteomics and phospho-proteomics in *Xenopus* oocytes from prophase I through the MII arrest of the unfertilized egg. The data contains protein abundance and phosphorylation sites of a large number set of proteins at different stages of oocyte maturation. The large sets of data are of high quality. In addition, the authors discussed several key pathways critical for the maturation. The data is very useful for researchers, not only researchers in *Xenopus* oocytes but also those in oocyte biology in other organisms.

Strengths:

The data of proteomics and phospho-proteomics in *Xenopus* oocyte maturation is very useful for future studies to understand molecular networks in oocyte maturation.

Weaknesses:

Although the authors offered molecular pathways of the phosphorylation in translation, protein degradation, cell cycle regulation, and chromosome segregation. The authors did not check the validity of the molecular pathways based on their proteomic data by experimentation. But this is not essential since this is a resource paper.

<https://doi.org/10.7554/eLife.104255.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

*The study aims to create a comprehensive repository about the changes in protein abundance and their modification during oocyte maturation in *Xenopus laevis*.*

Strengths:

The results contribute meaningfully to the field.

Weaknesses:

The manuscript could have benefitted from more comprehensive analyses and clearer writing. Nonetheless, the key findings are robust and offer a valuable resource for the scientific community.

We would like to thank the reviewer for his/her positive feedback on our article. The public review points out that "The manuscript could have benefitted from more comprehensive analyses and clearer writing." We have rewritten several sections and provided more detailed explanations of the analysis and interpretation of some data (see below for details). We have also followed all of the reviewer's recommendations, some of which specifically highlighted areas lacking clarity. We would also like to thank the reviewer for pointing out some errors, for which we apologize, and which have now been corrected. We sincerely appreciate the reviewer's thorough work, as it has greatly enhanced the clarity and precision of the manuscript.

Reviewer #2 (Public review):

Summary:

*The authors analyzed *Xenopus* oocytes at different stages of meiosis using quantitative phosphoproteomics. Their advanced methods and analyses revealed changes in protein abundances and phosphorylation states to an unprecedented depth and quantitative detail. In the manuscript they provide an excellent interpretation of these findings putting them in the context of past literature in *Xenopus* as well as in other model systems.*

Strengths:

High quality data, careful and detailed analysis, outstanding interpretation in the context of the large body of the literature.

Weaknesses:

Merely a resource, none of the findings are tested in functional experiments.

*I am very impressed by the quality of the data and the careful and detailed interpretation of the findings. In this form the manuscript will be an excellent resource to the cell division community in general, and it presents a very large number of hypotheses that can be tested in future experiments. *Xenopus* has been and still is a popular and powerful model system that led to critical discoveries around countless cellular processes, including the spindle, nuclear envelope, translational regulation, just to name a few. This also includes a huge body of literature on the cell cycle describing its phosphoregulation. It is indeed somewhat frustrating to see that these earlier studies using phosphomutants and phospho-antibodies were just scratching the surface. The phosphoproteomics analysis presented here reveals much more extensive and much more dynamic changes in phosphorylation states. Thereby, in my opinion, this manuscript opens a completely new chapter in this line of research, setting the stage for more systematic future studies.*

We thank the reviewer for his/her extremely positive comments. The public review points out that "none of the findings are tested in functional experiments." This is entirely accurate. We focused our work on obtaining the highest quality proteomic and phosphoproteomic data possible, and then sought to highlight these data by connecting them with existing functional data from the literature. This approach has opened up research avenues with enormous, previously unforeseen potential, in a wide range of biological fields (cell cycle, meiosis, oogenesis, embryonic development, cell biology, cellular physiology, signaling, evolution, etc.). We chose not to delay publication by experimentally investigating the narrow area in which we are specialists (meiotic maturation), while our data offer a vast array of research opportunities across various fields. Our goal was, therefore, to present this extensive dataset as a resource for different scientific communities, who can explore their specific biological questions using our data. This is why we submitted our article to the "Repository" section of eLife. Nevertheless, in the context of the comparative analysis of the mouse and *Xenopus* phosphoproteomes performed at the reviewer's request, we felt it was important to complement this new section with functional experiments that not only validate the proteomic data but also provide new insights into certain proteins and their regulation by Cdk1 (new paragraph lines 824-860 and new Figure 9).

We are also grateful to the reviewer for the recommendation to improve the manuscript by including more comparisons between our *Xenopus* data and those from other systems. We have followed this suggestion (see below), which has significantly enriched the article (new paragraph lines 824-860 and new Figure 9).

Reviewer #3 (Public review):**Summary:**

*The authors performed time-resolved proteomics and phospho-proteomics in *Xenopus* oocytes from prophase I through the MII arrest of the unfertilized egg. The data contains protein abundance and phosphorylation sites of a large number set of proteins at different stages of oocyte maturation. The large sets of the data are of high quality. In addition, the authors discussed several key pathways critical for the maturation. The data is very useful for the researchers not only researchers in *Xenopus* oocytes but also those in oocyte biology in other organisms.*

Strengths:

The data of proteomics and phospho-proteomics in Xenopus oocyte maturation is very useful for future studies to understand molecular networks in oocyte maturation.

Weaknesses:

Although the authors offered molecular pathways of the phosphorylation in the translation, protein degradation, cell cycle regulation, and chromosome segregation. The author did not check the validity of the molecular pathways based on their proteomic data by the experimentation.

We thank the reviewer for his/her positive comments. The public review points out that "The author did not check the validity of the molecular pathways based on their proteomic data by the experimentation." This is entirely accurate. We focused our work on obtaining the highest quality proteomic and phosphoproteomic data possible, and then sought to highlight these data by connecting them with existing functional data from the literature. This approach has opened up research avenues with enormous, previously unforeseen potential, in a wide range of biological fields (cell cycle, meiosis, oogenesis, embryonic development, cell biology, cellular physiology, signaling, evolution, etc.). We chose not to delay publication by experimentally investigating the very narrow area in which we are specialists (meiotic maturation), while our data offer a vast array of research opportunities across various fields. Our goal was, therefore, to present this extensive dataset as a resource for different scientific communities, who can explore their specific biological questions using our data. This is why we submitted our article to the "Repository" section of eLife. Nevertheless, in the context of the comparative analysis of the mouse and *Xenopus* phosphoproteomes performed at the reviewer's request, we felt it was important to complement this new section with functional experiments that not only validate the proteomic data but also provide new insights into certain proteins and their regulation by Cdk1 (new paragraph lines 824-860 and new Figure 9).

We have also followed all of the reviewer's recommendations and thank him/her, as the suggestions have significantly enhanced the manuscript.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Fig. 1 -> In the Figure legend "mPR β " is called "mPRb". In the Figure, it is indicated that PKA substrates are always activated by the phosphorylation. As the relevant substrates and the mode-of-action of the Arpp19 phosphorylation are not clear at the moment, this seems to be preliminary. It could for example also be conceivable that PKA phosphorylation inhibits a translation activator. In addition, the PG-dependent translation of RINGO/Speedy should be included in the model.

We fully agree with the reviewer. PKA substrates can either be activators of the Cdk1 activation pathway, which are inhibited by phosphorylation by PKA, or repressors of the same pathway, which are activated by phosphorylation by PKA. This is now illustrated in the new Fig. 1. In addition, we have also included RINGO/Speedy in the model and in the text (lines 78-79) and corrected "mPRb" in the legend.

(2) Lane 51-52 -> it is questionable if the meiotic divisions can be called "embryonic processes"

We agree with the reviewer comment, and we have removed the word "embryonic".

(3) Lane 53 and lane 106-107 -> recent data have indicated that transcription already starts during cell cycle 12 and 13 in most cells (e.g. Blitz and Cho: Control of zygotic genome activation in *Xenopus* (2021))

We apologize for this mistake. The text has been corrected and the reference added (lines 53 and 107).

(4) Lane 61-62 -> "MI" and "MII" are given as abbreviation for "first and second meiotic spindle"

The text has been clarified to explain that MI is referred to metaphase I and MII stands for metaphase II (lines 61-64).

(%) Lane 131-132 -> "single-cell" is mentioned redundantly in this sentence.

The sentence has been corrected (lines 131-132).

(6) Fig. 2B -> it is not explained what is plotted as "Average levels" on the x-Axis. Is it the average of expression over all samples or at a given time point? Are the values given as a concentration or are the values normalized? If so, how were they normalized?

We agree with the reviewer comment that "Average levels" may have been unclear. In the new Fig. 2B, we have re-plotted the graph using the average protein concentration during meiosis, measured as described in the Methods section.

(7) In Fig. 2-supplement 3E -> from the descriptions it is not entirely clear to me what the difference to the data in Fig. 2B is?

We thank the reviewer for his/her question regarding the relationship between the data in Fig. 2B and Fig. 2-supplement 3E. We confirm that the raw data visualized in Fig. 2-supplement 3E are the same as those in Fig. 2B. However, in Fig. 2-supplement 3E, the data are color-coded differently to highlight the number of proteins whose concentrations change during meiotic divisions, based on the threshold adopted. The legend of Fig. 2-supplement 3E has been modified to clarify this point.

(8) Lane 225-226 -> *Kifc1* is a minus-end directed motor

This mistake has been corrected (lines 232-233).

(9) Lane 271 -> *Serbp1*, here mentioned to be involved in stabilization of mRNAs, has also been implicated in the regulation of ribosomes (e.g. Leesch et al. 2023). Regarding the overall topic of this manuscript, this could be mentioned as well.

We agree with the referee that the important role of *Serbp1* in the control of ribosome hibernation needs to be mentioned. We have included this point in the revised manuscript together with the reference (lines 277-279).

(10) Lane 360-363 -> it is mentioned that *APPL1* and *Akt2* act "to induce meiosis". Furthermore, in the Nader et al. 2020 paper, *Akt2* phosphorylation is reported to happen within 30min after PG treatment. In the present work, they only seem to get phosphorylated when *Cdk1* is activated. Is there an explanation for this discrepancy?

Indeed, Nader et al. (2020) indicate that Akt2 is phosphorylated on Ser473 (actually, they should have mentioned Ser474, which is the phosphorylated residue on Akt2; Ser473 corresponds to the numbering of Akt1) between 5 and 30 minutes post-Pg, which supports their hypothesis of an early role for this kinase. However, these conclusions should be taken with caution, considering that their functional experiment using antisense against Akt2 depletes only 25% of the protein, the antibody used to visualize Akt2 phosphorylation also recognizes phosphorylated Akt1 and Akt3, and they did not analyze phosphorylation of the protein after 30 minutes. Therefore, we cannot determine whether the level observed at 30 minutes represents a maximum or if it is just the onset of the phosphorylation that peaks later, possibly after activation of Cdk1, for example.

Regarding our measurements: we clearly observe phosphorylation of Akt2 following Cdk1 activation on Ser131. We did not detect Akt2 phosphorylation on Ser474, but since our measurements started 1 hour post-Pg, this protein may have returned to a dephosphorylated state on Ser474.

Therefore, the observations of Nader et al. and ours involve different residues and different phosphorylation kinetics, Nader et al. limiting their analysis to the first 30 minutes, whereas we started at 1 hour.

We have revised the manuscript text to make these aspects clearer (lines 387-392).

| (11) Fig. 3B -> it could be made clearer in the Figure that all these sites belong to class I

A title "Class I proteins" has been added in Fig. 3B to clarify it.

| (12) Lane 433-434 -> the authors write that the proteomic data of this study confirm that PATL1 is accumulating during meiotic maturation. However, in Fig. 2B PATL1 is not among the significantly enriched proteins.

We apologize for this error. Indeed, PATL1 protein is not significantly enriched. The text has been corrected (lines 461-465).

| (13) Fig. 4B -> Zar2 is color-coded to increase in abundance. This is clearly different to published results and what is shown in Fig. 2B of this manuscript.

Indeed, our dataset shows that the quantity of Zar2 decreases. This does not appear anymore in Figure 2B since Zar2 average concentration cannot be estimated. We made an error in the color coding, which has now been corrected in Figure 4B.

| (14) Lane 442-444 -> it might be worth mentioning that the interaction between CPEB1 and Maskin, and thus probably its role in regulation of translation, could not be reproduced in other studies (Minshall et al.: CPEB interacts with an ovary-specific eIF4E and 4E-T in early *Xenopus* oocytes (2007) or Duran-Arque et al.: Comparative analyses of vertebrate CPEB proteins define two subfamilies with coordinated yet distinct functions in post-transcriptional gene regulation (2022)).

This clarification is now mentioned in the text, supported by the two references that have been added (lines 471-477).

| (15) Lane 483-485 -> The meaning of these sentences is not entirely clear to me. What exactly is the similarity with the function of Emi1? What does "...binding of Cyclin B1..." mean (binding to which other protein?). What is the similarity between Emi1 and CPEB1/BTG4, both of which are regulators of mRNA stability/polyadenylation?

We apologize if these sentences were unclear. Our intention was to emphasize the central role of ubiquitin ligases in regulating multiple events during meiotic divisions. We used SCF^{βTrCP}, a wellstudied ubiquitin ligase in *Xenopus* and mouse oocytes during meiosis, as an example. SCF^{βTrCP} regulates the degradation of several substrates, including Emi1, Emi2, CPEB1, and Btg4, whose degradation or stabilization is essential for the proper progression of meiosis. Lastly, we highlighted that these regulatory processes, mediated by protein degradation, may be conserved in mitosis, as for example the destruction of Emi1. We have rewritten this paragraph for clarity (lines 513-518).

(16) Lane 521-522 and 572-573 -> the authors write that Myt1 was not detected in their proteome. However, in Fig. 6A they list "pkmyt1" as a class II protein. On Xenbase, "pkmyt1" is the Cdk1 kinase, "Myt1" is a transcription factor, so the authors might have been looking for the wrong protein.

We thank the reviewer for this accurate observation. We have modified the text to correct this error (lines 554 and 607).

(17) Lane 564-565 -> The authors state that Cdk1 activity can be measured by analyzing Cdc27 S428 phosphorylation. However, in vivo the net phosphorylation of a site is always depending on the relevant kinase and phosphatase activities. As S428 is a Cdk1 site, it is not unlikely that it is dephosphorylated by PP2A-B55, which by itself is under the control of Cdk1. Do the authors have direct evidence that the change in phosphorylation of S428 can only be attributed to the changes in Cdk1 activity?

There is evidence in the literature that Cdc27 is dephosphorylated by PP2A (Torres et al., 2010). In *Xenopus* oocytes, PP2A activity is high during prophase (Lemonnier et al., 2021) and decreases at the time of Cdk1 activation, mediated by the Greatwall-ENSA/Arpp19 system, remaining low until MII (Labbé et al., 2021). Therefore, the period where fluctuations in Cdk1 activity are difficult to assess, from NEBD to MII, corresponds to a phase of inhibited PP2A activity. As a result, the phosphorylation level of Cdc27 reflects primarily the activity of Cdk1. We have added this clarification in the text (lines 597-600).

(18) Fig. 7C and 7D -> in 7C, for Nup35/Nup53 there is a phospho-peptide GIMEVRS(60)PPLHSGG. In Fig. 7D phosphorylation of GVMEMRS(59)PLFSGG is analyzed. Is this the same phosphosite/region of Nup35/Nup53? How can there be a slightly different version of the same peptide in one protein? Are these the L- and S-version of Nup35/Nup53? It is also very surprising that the two phosphosites belong to different classes, class III and class II, respectively.

We thank the reviewer for this observation. The peptides GIMEVRS(60)PPLHSGG and GVMEMRS(59)PLFSGG correspond to the same phosphorylation site in the L and S versions of *Xenopus laevis* Nup35, respectively. The L version peptide was classified as Class III, while the S version was not assigned to any class due to its high phosphorylation level in prophase, which prevented it from meeting the log₂ fold-change threshold of 1 required by our analysis to detect significant differences.

(19) Table 1 -> second last column is headed "Whur, 2014"

The typo has been corrected.

(20) Fig. 8 -> Why are all the traces starting at t=1h after PG?

The labeling of the graphs in Fig. 8 has been corrected, and the traces now begin at t0.

(21) Lane 754 -> *Although a minority, there are also some minus-end directed kinesins, e.g. Kifc1*

We agree with the reviewer. We should have mentioned that, in addition to dyneins, some kinesins are minus-end directed motors, especially since one of them, Kifc1, is regulated at the level of its accumulation. We have rephrased the relevant sentences to incorporate this observation (lines 790-793).

(22) Section "Assembly of microtubule spindles and microtubule dynamics" -> *Although this section clearly has a strong focus on phosphorylation, it might be worth mentioning again that many regulators of the microtubule spindle, e.g. TXP2, are among the upregulated proteins in Fig. 2B/C*

We have already discussed that the protein levels of certain key regulators of the mitotic spindle (Tpx2, PRC1, SSX2IP, Kif11/Eg5 among others) are subject to control during meiotic maturation in a previous chapter "Protein accumulation: the machinery of cell division and DNA replication" (lines 230-239). We agree with the reviewer that this important observation can be mentioned again at the beginning of this chapter on phosphorylation control. We have added a sentence regarding this at the start of the paragraph (lines 774-775).

Reviewer #2 (Recommendations for the authors):

While I find the manuscript excellent and detailed already in its current form, I would appreciate including even more comparisons to other systems. In particular, a similar phosphoproteomics experiment has been performed in starfish oocytes undergoing meiosis (Swartz et al, eLife, 2021), and there are several studies on mitosis of diverse mammalian cells. It would be very exciting to see to what extent changes are conserved.

We thank the reviewer for this recommendation, which we have attempted to follow. We have matched our dataset of mass spectrometry using the the phosphor-occupancy_matlab package, available as part of our code repository (<https://github.com/elizabeth-van-itallie>) previously described in (Van Itallie et al, 2025). Unfortunately, we were unable to match our dataset with the data from Swartz et al. (2021) on starfish oocyte due to the low sequence conservation. However, we have compared our dataset with the dataset from Sun et al. (2024) on mouse oocyte maturation. We identified a total of 408 conserved phosphorylation sites, which mapped to 320 proteins in *Xenopus* and 277 in mice (refer to a new paragraph: lines 824-860, new Figure 9, Methods: lines 1011-1032 and 1060-1065, and Appendix 7). The phosphorylation patterns during meiosis showed a significant crossspecies correlation (Pearson $r = 0.39$, $p < 0.0001$; see new Figure 9A), demonstrating the evolutionary conservation of phosphoproteomic regulation. Important phosphorylation events, including Plk1 at T201, Gwl at S467, and Erk2 at T188, were upregulated in both species, in line with the activation of the Cdk1 and MAPK signaling cascades (Figure 6B, new Figure 9A-B). We validated several of these phosphorylation sites by western blotting and demonstrated their dependency on Cdk1 activation (new Figure 9C). Together, these findings reinforce the notion that fundamental phospho-regulatory pathways are conserved during oocyte maturation in vertebrates.

Reviewer #3 (Recommendations for the authors):

(1) Page 6, the first paragraph of Results section: *Please describe the method on how the authors measured and quantified the proteomes in different stages of Xenopus oocyte maturation briefly. Without the experimental design, it is very hard to evaluate the results in the following paragraphs.*

As requested by the reviewer, we added a few sentences describing the method of proteomics and phosphoproteomics measurements in oocytes resuming meiosis (lines 151-158).

(2) In the phospho-proteome, it is better to classify the amino acids for the phosphorylation such as Ser, Thr, and Tyr. Particularly how many tyrosine phosphorylations are in the list.

Our phosphosites dataset contains 80% Ser, 19.9% Thr, and 0.01% Tyr. Phospho-Tyr are slightly less abundant than what has been described in the literature (in most cells “roughly 85-90% of protein phosphorylation happens on Ser, ~10% on Thr, and less than 0.05% on Tyr” after Sharma et al., 2014. The same observation was made regarding the distribution of phosphorylated amino acids in mouse oocytes, where phospho-Tyr abundance is relatively diminished in oocytes compared to mouse organs (Sun et al., 2024). These observations are now reported in the manuscript (lines 309-313).

(3) In class II (Figure 3), when Cdk1 (line 326) is a major kinase, how many phosphorylation sites are a target of Cdk1 (with the Cdk1-motif)? Moreover, do the authors find any other consensus sequences for the phosphorylation? Those are either known or unknown. This information would be useful for the readers.

We thank the reviewer for this valuable comment. To address it, we used the kinase prediction server (<https://kinase-library.phosphosite.org/kinase-library/score-site>) to analyze Class II phosphosites. These new results are mentioned in lines 340-349 and illustrated in a new Figure (Figure 3—figure supplement 1A). We identified 303 sites predicted to be phosphorylated by Cdk1. Of these, 166 were also predicted as Erk1/2 targets, reflecting the similarity between Cdk1 and Erk1/2 consensus motifs.

Cdk1 substrate phosphorylation is governed by more than just the presence of a consensus sequence. In addition to its preference for the (S/T)P×(K/R) motif, Cdk1/cyclin complexes achieve specificity through docking interactions with short linear motifs (SLiMs) recognized by the cyclin subunit (as LxF motifs)(Loog & Morgan, 2005), and via the Cdk-binding subunits Cks1 or Cks2, which interact with phosphorylated threonine residues in primed substrates (Örd et al, 2019). These mechanisms promote processive multisite phosphorylation and allow Cdk1 to target substrates even at non-canonical sites. Our motif-based analysis captures only part of this complexity and may underestimate the number of true Cdk1 targets.

To further explore kinase involvement across phosphosite classes, we extended the analysis to all clusters and identified the most enriched kinase predictions for each (lines 360-365, new Figure 3—figure supplement 1B). In Class II, the most enriched kinases included Cdk1, Erk2, and Plk1, supporting the conclusions derived from the identification of the phosphosites of this Class. But others such as Cdk2, Cdk3, Cdk5, Cdk16, KIS, JNK1, and JNK3 were also identified.

(4) Figure 3B: Why do the authors show this kind of Table only for Class I, not Classes II-V? It would be informative to show candidate proteins in other classes.

We chose to present the candidate proteins from Class I in a table format because the number of phosphosites (136) was too small to allow a meaningful Gene Ontology (GO) enrichment analysis. Therefore, we manually curated the data and highlighted proteins whose Class I phosphosites are associated with specific biological processes. For Classes II–V, the higher number of phosphosites allowed us to perform GO enrichment analyses. Since several of the enriched processes were shared across different classes, and some proteins have phosphosites in multiple classes, we opted to organize the results by biological processes rather than by class. We agree with the reviewer that it is indeed valuable to highlight

interesting proteins with Class II–V phosphosites. We have done so in Figures 4 through 8, using graphical representations instead of tables, in order to make the data more accessible and avoid long tables. Additionally, the Supplementary Figures provide detailed phosphorylation trends for many of the proteins discussed in the main figures.

*(5) It would be nice if the authors compare this phospho-proteome in *Xenopus* oocyte maturation with that in mouse oocyte maturation (Sun et al. 2024) in terms of evolutionary conservation of the phospho-proteomes.*

We thank the reviewer for this suggestion. As now detailed in the manuscript, we compared our *Xenopus* phosphoproteome with the dataset from Sun et al. (2024) on mouse oocyte maturation using the `phospho_occupancy_matlab` package, available as part of our code repository (<https://github.com/elizabeth-van-itallie>) previously described in (Van Itallie et al, 2025). We identified 408 conserved phosphorylation sites corresponding to 320 *Xenopus* and 277 mouse proteins (see new paragraph: lines 824-860, new Figure 9, Methods: lines 1011-1032 and 1060-1065, and Appendix 7). Phosphorylation dynamics across meiosis were significantly correlated between the species (Pearson $r = 0.39$, $p < 0.0001$; new Figure 9A), highlighting evolutionary conservation of the phosphoproteomes. Key phosphorylation events such as Plk1 at T201, Gwl at S467, and Erk2 at T188 increased in both species, consistent with activation of the Cdk1 and MAPK pathways (Figure 6B, new Figure 9A–B). We validated experimentally several of these phosphorylation sites by western blot (Erk2, Plk1, Fak1 and Akts1) and demonstrated their dependency on Cdk1 activation (new Figure 9C). Together, these new findings support the conservation of key phospho-regulatory mechanisms across vertebrate oocyte maturation.

Minor points:

(1) Reference lists: Please add Sun et al (2024) shown in line 115.

This important reference has been added (lines 115, 134, 313 and 826).

(2) Figure 1, red arrows for the inhibition: This should be "T" shape for a better understanding of these complicated pathways.

We agree with the reviewer's remark, and we have modified Figure 1.

*(3) Line 236-238: The authors referred to the absence of Cdc6 in oocyte maturation in *Xenopus*. However, Figure 2C shows that Cdc6 belongs to a list of accumulating proteins with Orc1 and Ocr2 etc. and the authors did not discuss this discrepancy in the text. Please clarify the claim.*

We apologize for the unclear wording in our text. The section of the manuscript regarding the pre-RC components may have been misleading. The text has been revised to clarify that Cdc6 was not detected in prophase-arrested oocytes by western blot and that it accumulates during meiotic maturation after MI, enabling oocytes to replicate DNA (lines 243-250).

(4) Line 306: Please add the link to phosphosite.org.

The link has been added (line 319).

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